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A SECTORAL TAXONOMY OF AI INTENSITY

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A sectoral taxonomy of AI intensity

Flavio Calvino, Hélène Dernis, Lea Samek and Antonio Ughi

This work proposes a sectoral taxonomy of AI intensity, outlining different dimensions that characterise the extent to which AI relates to the activity of economic sectors. Focusing on AI human capital, AI innovation, exposure to and use of AI, the taxonomy provides a novel multifaceted perspective and reveals significant heterogeneity across sectors and indicators. While some sectors, such as IT services, score high along all the dimensions considered, others, such as Pharmaceuticals, exhibit more considerable heterogeneity (high AI human capital but low AI innovation). The taxonomy can be a useful tool for future policy-relevant analyses aimed at exploring empirically the role of AI and the implications of its diffusion.

Keywords: Artificial Intelligence, Human Capital, Innovation

JEL codes: C81, O33, O34, J24

Résumé

Ce travail propose une taxonomie sectorielle de l'intensité de l'IA, par le biais de plusieurs dimensions caractérisant les liens entre l'IA et l'activité des secteurs économiques. Axée sur le capital humain, l'innovation, l'exposition et l'utilisation de l'IA, la taxonomie fait émerger un panorama à multiples facettes et révèle une hétérogénéité significative entre les secteurs et les indicateurs. Alors que certains secteurs, tels les services informatiques, obtiennent des résultats élevés pour toutes les dimensions considérées, d'autres, comme l'industrie pharmaceutique, présentent une hétérogénéité plus forte (capital humain en IA élevé mais innovation en IA faible). La taxonomie fournit un outil qui peut être utile pour de futures analyses politiques visant à explorer le rôle de l'IA et les implications de sa diffusion de manière empirique.

Kurzfassung

In dieser Arbeit wird eine sektorale Taxonomie der KI-Intensität vorgeschlagen, die verschiedene Dimensionen beschreibt, welche das Ausmaß der Verbindung zwischen KI und den Tätigkeiten von Wirtschaftssektoren charakterisieren. Mit dem Schwerpunkt auf KI-Humankapital, KI-Innovation, KI-Exposition und KI-Nutzung bietet die Taxonomie eine neuartige, vielschichtige Perspektive und zeigt eine erhebliche Heterogenität zwischen Sektoren und Indikatoren auf. Während einige Sektoren, wie z. B. IT-Dienstleistungen, in allen betrachteten Dimensionen hohe Werte erreichen, weisen andere, wie z. B. die Pharmaindustrie, größere Unterschiede auf (hohes KI-Humankapital, aber geringe KI-Innovation). Die Taxonomie kann ein nützliches Instrument für künftige politikrelevante Analysen sein, die empirisch die Rolle der KI und die Auswirkungen ihrer Verbreitung untersuchen.

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Executive summary

Artificial Intelligence (AI) has the potential to significantly reshape economies, permeate many occupations and sectors, and affect the way people work and firms organise their production. For policy makers to effectively assess the impacts of AI and capitalise on the opportunities that the digital transformation can bring about, it is important to improve the understanding of how AI diffuses and how it affects different economic activities. This requires the availability of relevant indicators at a granular level, which can be related to policy-relevant economic or social outcomes. In this context, this work proposes a sectoral taxonomy of AI intensity, which appears particularly relevant from a policy perspective. Focusing on different facets, as measured by the different indicators, such an operational tool can be applied in a range of policy-relevant analyses, at the microeconomic or sectoral level, to further explore empirically the role of AI for different economic or social outcomes. It builds on and significantly extends the existing taxonomy proposed by Calvino et al. (2018[1]), which classified the digital intensity of sectors and has been cited in over 350 subsequent analyses to assess the role of digitalisation, for instance, for market power, firm dynamics, or skill endowment.

The proposed taxonomy outlines different facets that characterise the extent to which AI relates to the activity of economic sectors. In particular, sectors differ in their demand for AI talent necessary to develop and maintain AI systems, the AI innovations that are developed as a result, their exposure to AI systems as well as their current use of AI. These facets are proxied with four indicators, calculated at the sectoral level and include: i) the share of AI-related online vacancies; ii) the share of AI-related patents; iii) a barrier-adjusted measure of exposure to AI; iv) the share of firms using AI. A novel adjustment to a previously proposed indicator of AI exposure aims at accounting for policy-relevant barriers constraining exposure and hence actual AI use, reflecting their impact on the potential of AI to affect tasks carried out by workers in each sector.

The analysis reveals significant heterogeneity across sectors and indicators. While some sectors, such as 'IT services', 'Media' or 'Telecommunications', score high along all the dimensions considered, others exhibit more considerable heterogeneity. For instance, 'Pharmaceuticals' show high AI human capital but low AI innovation, possibly because they rely more on AI applications used and maintained by AI talent than on developing novel, patentable AI based solutions. The 'Food products', 'Textiles & apparel' and 'Wood & paper' sectors in manufacturing, along with 'Construction', result among the least AI intensive sectors for most indicators considered.

Examining sectoral rankings of barrier-adjusted AI exposure alongside actual AI use could also offer insights about the pace of AI diffusion, as it may allow to further relate the potential and actual use of AI based on the relative positions of sectors. For example, the manufacturing of 'Computer and electronics' sector ranks highly in barrier-adjusted AI exposure but lower in actual use, possibly suggesting a slower pace of AI diffusion compared to sectors where the relative position in barrier-adjusted exposure is lower than in actual use, such as 'Chemicals' or 'Transport equipment'.

A possible summary AI intensity indicator, based on information resulting from the four indicators, shows that highly AI intensive sectors include 'Manufacturing of Computer & electronics', 'Media', 'Telecommunications', 'IT services', 'Finance & insurance', 'Legal & accounting' and 'Scientific R&D'. Low

AI intensity is found in sectors such as ‘Food products’, ‘Textiles & apparel’, ‘Wood & paper’, ‘Rubber, plastics, minerals’, ‘Metal products’, ‘Construction’ as well as ‘Hotels & food services’. The remaining manufacturing sectors as well as ‘Wholesale & retail’, ‘Transportation & storage’, ‘Real estate’, ‘Other business services’ and ‘Administrative & support services’ fall in the medium category.

Relating this measure to the existing OECD taxonomy of digital intensity suggests that, while AI diffusion appears to rely on the digital transformation, there are also relevant sectoral differences. This work therefore complements other taxonomies, such as those based on R&D or digital intensities.

Synthèse

L'intelligence artificielle (IA) a le potentiel de remodeler considérablement les économies, d'imprégner de nombreuses professions et de nombreux secteurs, et de transformer la façon dont les gens travaillent et dont les entreprises organisent leur production. Pour que les décideurs politiques puissent évaluer efficacement les impacts de l'IA et capitaliser sur les opportunités que la transformation numérique peut apporter, il est important de mieux comprendre la manière dont l'IA se diffuse et comment elle touche les différentes activités économiques. Pour ce faire, il convient de disposer d'indicateurs pertinents à un niveau granulaire, en lien à des résultats économiques ou sociaux pertinents pour les politiques. Dans ce contexte, ce travail propose une taxonomie sectorielle de l'intensité de l'IA, qui semble pertinente d'un point de vue politique. En se concentrant sur les différentes facettes offertes par différents indicateurs, un tel outil opérationnel peut être appliqué dans nombre d'analyses politiques, au niveau microéconomique ou sectoriel, afin d'explorer de manière empirique le rôle de l'IA dans une série de résultats économiques ou sociaux. Il s'appuie sur la taxonomie proposée par Calvino et al. (2018[1]), qui classe l'intensité numérique des secteurs, et la prolonge. Celle-ci a été citée dans plus de 350 analyses ultérieures pour évaluer le rôle de la numérisation, par exemple, sur le pouvoir de marché, la dynamique des entreprises ou la dotation en compétences.

La taxonomie proposée découvre différentes facettes caractérisant les liens entre l'IA et l'activité des secteurs économiques. En particulier, les secteurs diffèrent dans leur demande de talents en IA nécessaires au développement et à la maintenance des systèmes d'IA, dans les innovations en IA qui en découlent, dans leur exposition aux systèmes d'IA ainsi que dans leur utilisation actuelle de l'IA. Ces facettes sont représentées par quatre indicateurs, calculés au niveau sectoriel et comprenant : i) la part des offres d'emploi en ligne liées à l'IA ; ii) la part des brevets liés à l'IA ; iii) une mesure de l'exposition à l'IA ajustée en fonction des obstacles ; iv) la part des entreprises utilisant l'IA. Un nouvel ajustement de l'indicateur d'exposition à l'IA précédemment défini vise à tenir compte des obstacles liés à la politique qui limitent l'exposition et donc l'utilisation réelle de l'IA, en reflétant leur impact sur le potentiel de l'IA à affecter les tâches effectuées par les travailleurs dans chaque secteur.

L'analyse révèle une hétérogénéité significative entre secteurs et entre indicateurs. Alors que certains secteurs, tels que les « services informatiques », les « médias » ou les « télécommunications », obtiennent des résultats élevés dans toutes les dimensions, d'autres sont plus hétérogènes. Par exemple, les « produits pharmaceutiques » affichent un capital humain en IA élevé mais une faible innovation en IA, peut-être parce qu'ils s'appuient davantage sur des applications d'IA utilisées et maintenues par des spécialistes de l'IA que sur le développement de solutions nouvelles et brevetables basées sur l'IA. Les secteurs manufacturiers des « produits alimentaires », du « textile et de l'habillement » et du « bois et papier », ainsi que la « construction », figurent parmi les secteurs les moins intensifs en IA pour la plupart des indicateurs considérés.

L'examen des classements sectoriels de l'exposition à l'IA corrigée des obstacles et de l'utilisation réelle de l'IA pourrait également fournir des indications sur le rythme de diffusion de l'IA, permettant de tisser un lien plus étroit entre l'utilisation potentielle et réelle de l'IA, fondé sur le positionnement relatif des secteurs. Par exemple, le secteur de la fabrication d'« ordinateurs et d'appareils électroniques » possède une forte exposition à l'IA corrigée des barrières, mais est moins bien classé en termes d'utilisation réelle. Ce résultat suggère un rythme de diffusion de l'IA plus lent que dans les secteurs où la position relative en termes d'exposition corrigée des barrières est plus faible qu'en termes d'utilisation réelle, tels que les « produits chimiques » ou le « matériel de transport ».

Un indicateur synthétique possible de l'intensité de l'IA, basé sur les quatre indicateurs, montre que les secteurs à forte intensité d'IA incluent la « Fabrication d'ordinateurs et d'électronique », les « Médias », les « Télécommunications », les « Services informatiques », la « Finance et assurance », le « Juridique et comptabilité » et la « Recherche et développement scientifique ». Une faible intensité d'IA est observée dans des secteurs tels que « Produits alimentaires », « Textile et habillement », « Bois et papier », « Caoutchouc, plastique, minéraux », « Produits métalliques », « Construction » ainsi que « Hôtellerie et restauration ». Les autres secteurs manufacturiers ainsi que les secteurs « Commerce de gros et de détail », « Transport et entreposage », « Immobilier », « Autres services aux entreprises » et « Services administratifs et de soutien » entrent dans la catégorie d'intensité d'IA moyenne.

La mise en relation de cette mesure avec la taxonomie existante de l'OCDE sur l'intensité numérique suggère que, si la diffusion de l'IA semble dépendre de la transformation numérique, il existe également des différences sectorielles. Ce travail complète donc d'autres taxonomies, telles que celles basées sur la R&D ou les intensités numériques.

Zusammenfassung

Künstliche Intelligenz (KI) hat das Potenzial, die Volkswirtschaften erheblich umzugestalten, viele Berufe und Sektoren zu durchdringen und die Art und Weise zu beeinflussen, wie Menschen arbeiten und Unternehmen ihre Produktion organisieren. Damit Entscheidungsträger die Auswirkungen der KI effektiv bewerten und die Chancen, die der digitale Wandel bieten kann, nutzen können, ist es wichtig, besser zu verstehen, wie sich KI verbreitet und wie sie verschiedene wirtschaftliche Aktivitäten beeinflusst. Hierfür ist die Verfügbarkeit relevanter Indikatoren auf granularer Ebene erforderlich, die mit politikrelevanten wirtschaftlichen oder sozialen Ergebnissen in Verbindung gebracht werden können. In diesem Zusammenhang schlägt diese Arbeit eine sektorale Taxonomie der KI-Intensität vor, die aus politischer Sicht von besonderer Relevanz erscheint. Durch den Fokus auf diverse Facetten, die durch die unterschiedlichen Indikatoren gemessen werden, kann ein solches operatives Instrument in einer Reihe von politikrelevanten Analysen auf mikroökonomischer oder sektoraler Ebene angewandt werden, um die Rolle der KI für verschiedene wirtschaftliche oder soziale Ergebnisse empirisch weiter zu untersuchen. Sie baut auf der bestehenden Taxonomie von Calvino et al. (2018[1]) auf und erweitert diese erheblich. Diese Taxonomie klassifiziert die digitale Intensität von Sektoren und wurde in mehr als 350 nachfolgenden Analysen zitiert, um beispielsweise die Rolle der Digitalisierung für Marktmacht, Unternehmensdynamik oder Fähigkeitsniveau zu bewerten.

Die vorgeschlagene Taxonomie beschreibt verschiedene Facetten, die das Ausmaß der Verbindung zwischen KI und den Tätigkeiten von Wirtschaftssektoren characterisieren. Insbesondere unterscheiden sich die Sektoren in ihrer Nachfrage nach KI-Talenten, die für die Entwicklung und Wartung von KI-Systemen erforderlich sind, in den KI-Innovationen, die daraus hervorgehen, ihrer Exposition gegenüber KI-Systemen sowie in ihrer aktuellen Nutzung von KI. Diese Facetten werden mit vier Indikatoren dargestellt, die auf sektoraler Ebene berechnet werden und Folgendes umfassen: i) den Anteil KI-bezogener Online-Stellenangebote; ii) den Anteil KI-bezogener Patente; iii) ein barrierebereinigtes Maß für die Exposition gegenüber KI; iv) den Anteil der Unternehmen, die KI nutzen. Eine neuartige Anpassung eines zuvor vorgeschlagenen Indikators für die KI-Exposition berücksichtigt politikrelevante Hindernisse, die die Exposition und damit die tatsächliche KI-Nutzung einschränken. Sie zeigt deren Auswirkungen auf das Potenzial der KI, die ausgeführten Aufgaben von Arbeitnehmern in den einzelnen Sektoren zu beeinflussen.

Die Analyse zeigt eine erhebliche Heterogenität zwischen den Sektoren und Indikatoren. Während einige Sektoren wie „IT-Dienstleistungen“, „Medien“ oder „Telekommunikation“ in allen betrachteten Dimensionen hohe Werte erzielen, weisen andere eine größere Heterogenität auf. Die Pharmaindustrie beispielsweise weist ein hohes KI-Humankapital, aber eine niedrige KI-Innovationsrate auf, was möglicherweise darauf zurückzuführen ist, dass sie sich mehr auf KI-Anwendungen stützt, die von KI-Talenten genutzt und gewartet werden, als auf die Entwicklung neuer, patentierbarer KI-basierter Lösungen. Die Sektoren „Lebensmittelprodukte“, „Textilien und Bekleidung“ sowie „Holz und Papier“ in der Produktion, zusammen mit dem „Baugewerbe“ gehören zu den am wenigsten KI-intensiven Sektoren für die meisten betrachteten Indikatoren.

Die Untersuchung der sektoralen Rangfolge der barrierebereinigten KI-Exposition im Vergleich zur tatsächlichen KI-Nutzung könnte auch Aufschluss über das Tempo der KI-Verbreitung geben, da sie es ermöglicht, das Potenzial und die tatsächliche Nutzung von KI anhand der relativen Positionen der Sektoren weiter zu bretrachten. Zum Beispiel liegt der Sektor „Computer und Elektronik“ in der KI-Exposition unter Berücksichtigung von Barrieren hoch, bei der tatsächlichen Nutzung jedoch niedriger. Dies könnte auf eine langsamere KI-Verbreitung im Vergleich zu Sektoren hinweisen, in denen die relative Position bei der barrierebereinigten Exposition niedriger ist als bei der tatsächlichen Nutzung, wie etwa bei „Chemie“ oder „Transportausrüstung“.

Ein möglicher zusammenfassender Indikator für die KI-Intensität, basierend auf den Informationen der vier Indikatoren, zeigt, dass zu den Sektoren mit hoher KI-Intensität die „Herstellung von Computern und Elektronik“, „Medien“, „Telekommunikation“, „IT-Dienstleistungen“, „Finanz- und Versicherungswesen“, „Rechts- und Steuerberatung“ sowie „Wissenschaftliche Forschung und Entwicklung“ (F&E) gehören. Geringe AI-Intensität findet sich in Sektoren wie „Lebensmittelprodukte“, „Textilien und Bekleidung“, „Holz und Papier“, „Gummi, Kunststoffe, Mineralien“, „Metallerzeugnisse“, „Baugewerbe“ sowie „Hotels und Gastronomie“. Die übrigen Produktionssektoren sowie „Groß- und Einzelhandel“, „Transport und Lagerung“, „Immobilien“, „Sonstige geschäftliche Dienstleistungen“ und „Verwaltung und unterstützende Dienstleistungen“ fallen in die mittlere Kategorie.

Ein Vergleich dieses Maßes mit der bestehenden OECD-Taxonomie der digitalen Intensität zeigt, dass die KI-Verbreitung zwar von der digitalen Transformation abhängig zu sein scheint, es aber auch relevante sektorale Unterschiede gibt. Diese Arbeit ergänzt daher andere Taxonomien, z. B. solche, die auf F&E oder digitalen Intensität basieren.

1 Introduction

Artificial Intelligence (AI) has the potential to significantly reshape economies, permeate many occupations and sectors and affect the way people work and firms organise their production. Therefore, it is often said to have transformative power with the potential to improve efficiency and innovative capacity. However, as it affects several dimensions of economic activities to different extents and in heterogeneous ways - but nonetheless at a fast pace - measuring its diffusion, the ways and the intensity with which it is adopted is inherently difficult.

For policy makers to effectively assess the impacts of AI and capitalise on the opportunities that the digital transformation can bring about, it is important to improve the understanding of how AI diffuses and affects different economic activities. Although this requires capturing the different facets of AI diffusion, due to limited data availability existing studies have often focused on only one facet at a time to proxy AI diffusion. However, a sector where many AI systems are used might not necessarily be characterised by a very high presence of AI innovation, and vice versa. Alternatively, a sector may have high exposure to AI because occupations in that sector tend to carry out tasks that can potentially be complemented or substituted by AI, but still face limited actual use of AI due to adoption barriers such as high implementation costs or ethical concerns.

There are ongoing initiatives with a more holistic view which leverage live data to inform researchers and support policy making. The OECD.AI Policy Observatory as well as the European Commission's AI Watch, for instance, provide insights across countries on trends in the fields of AI capabilities, AI research, AI human capital and standards for trustworthy AI, to name just a few, thereby offering a one-stop hub for AI policy resources. The AI Index developed by the Institute for Human-Centred AI at Stanford University is another initiative tracking also areas like public opinion on current and potential future impacts of AI and policy measures taken to stimulate AI innovation (Maslej et al., 2024[2]).

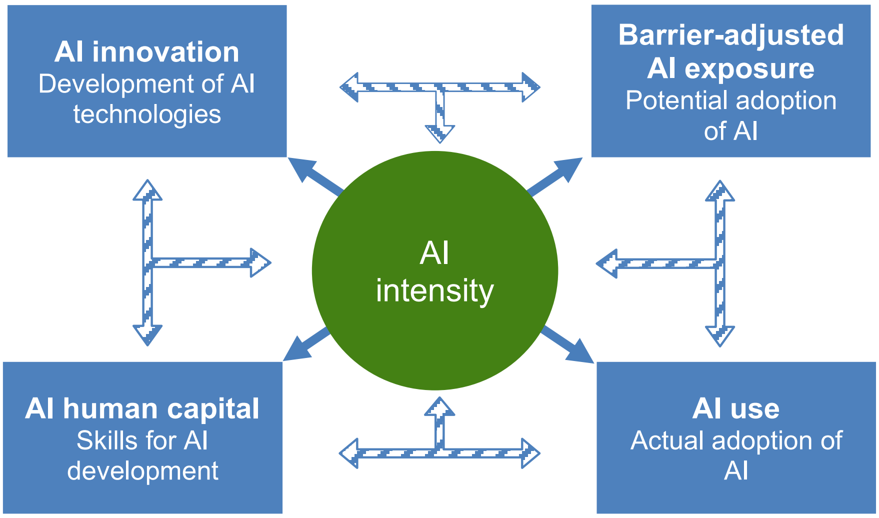
However, assessing how the spread of AI relates to relevant economic outcomes, e.g., how AI use translates into productivity gains or whether these gains emerge at the expense of workers by substituting certain tasks, requires indicators at granular level. Furthermore, depending on the purpose of analysis, some proxies of AI intensity may be more appropriate than others. In this context, the present study proposes a taxonomy that classifies sectors according to their AI intensity, accounting for key facets related to AI diffusion while recognising its heterogeneity across sectors. This work builds on and significantly extends an existing taxonomy proposed by Calvino et al. (2018[1]), who classified the digital intensity of sectors along several dimensions. Their taxonomy, which has been cited in over 350 subsequent analyses at the time of writing, has been applied to assess the role of digitalisation, for instance, for market power, firm dynamics, or skill endowment. By focusing on a key technological advancement of the digital transformation in recent years, namely AI, the taxonomy here further considers different facets of AI intensity:

- AI human capital,
- AI innovation,
- Barrier-adjusted AI exposure, and
- AI use.

By adopting a multifaceted perspective, the taxonomy accounts for some of the key dimensions that AI diffusion involves and recognises that sectors differ in their demand for AI talent necessary to develop and maintain AI-related technologies, the AI-related technologies that are developed as a result, their potential as well as their actual adoption. After all, AI does not only entail investing in and developing the most advanced algorithms, but also embedding them in production with the appropriate human capital and considering barriers to adoption (Figure 1.1).

Figure 1.1. Dimensions representing AI intensity

Interactions of different facets and indicators used in the sectoral taxonomy



Source: authors' elaboration.

AI human capital is proxied by AI skill demand in online vacancy data provided by Lightcast (2024[3]), AI innovation through AI-related patents derived from the latest available version of the European Patent Office (EPO) Worldwide Patent Statistical Database (i.e., PATSTAT database), maintained within the OECD STI Micro-data Lab, and AI use using publicly available data collected by National Statistical Offices (NSOs) based on the annual Eurostat survey on ICT usage and e-commerce in enterprises. Barrier-adjusted AI exposure is constructed by building on the AI industry exposure measure developed by Felten et al. (2021[4]). The measure is extended to further account for different barriers to adoption of AI, namely high costs and low maturity of the technology, operational barriers and lacking skills, as well as regulatory and ethical concerns.

To represent AI intensity at the sector level along these different dimensions and enhance comparability, coverage of countries, sectors and time frame are aligned to the best extent possible but still vary depending on the available information for each indicator. Given its sectoral focus as well as considering measurement challenges, the taxonomy does not fully capture within-sector heterogeneity. However, when possible, the analysis explores the role of cross-country differences within sectors, as well as time variation. Despite absolute differences between the countries considered, sectoral rankings within countries tend to remain rather stable. However, the taxonomy only focuses on (selected) OECD Member countries, which are likely to differ from developing or emerging economies.[1]

To help reduce the multidimensionality of the individual indicators considered, which are necessary to account for the complexity characterising AI diffusion, an overall summary indicator is also reported in this paper. While it may serve as an operational tool to support evidence-based policy making, it compares rather than quantifies AI intensity of sectors and masks some of the heterogeneity observed. Therefore, users may want to carefully consider the purpose of their analysis and consider employing the indicators individually rather than relying on the summary classification.

The remainder of the paper is structured as follows: Section 2 introduces each facet underlying the AI intensity taxonomy, discusses their rationale, explains the methodologies used to construct the indicators, and provides descriptive statistics, focusing on their variation along relevant dimensions. Section 3 proposes a sectoral taxonomy of AI intensity based on the different indicators, examining similarities and differences across sectors. It also discusses the possibility of reducing multidimensionality by presenting an overall summary indicator and discusses similarities and differences with respect to the taxonomy of digital-intensive sectors. The final section provides some concluding remarks and discusses possible next steps of analysis.

2 Key indicators of AI intensity

With AI affecting different dimensions of economic activities, one single indicator can hardly capture AI intensity comprehensively, especially because different sectors of activity are related to and affected by AI systems in heterogeneous ways. For instance, a sector may use many different AI systems but may not necessarily be characterised by a very high presence of AI innovation, and vice versa. Very recently, measures of heterogenous AI exposure across sectors, based on the tasks that occupations in different sectors carry out, gained considerable attention, also because such tasks can potentially be substituted or complemented by AI. However, being exposed to AI does not completely align with the current potential to use AI, notably because sectors may still face significant barriers to adoption, which can range from resource limitations to ethical considerations. In this context, this taxonomy aims at representing AI intensity along several dimensions, also capturing these nuances, by focusing on AI human capital, AI innovation, barrier-adjusted AI exposure and AI use and proposing relevant indicators to proxy each facet. This section discusses the rationale of the indicators and how they were constructed. Finally, it presents observable heterogeneity across sectors, also commenting on different dimensions, including country and time, when data availability allows.[2]

Although focusing on heterogeneous facets of AI intensity provides valuable information, the different indicators' availability and quality varies across data sources. Therefore, to enhance comparability across these dimensions, coverage of countries, sectors and time frame are aligned to the best extent possible. To this end, the following concerns have been addressed:

Sectoral coverage: Available data rely on different sectoral classifications and are often aggregated at different levels across countries and data sources. Moreover, coverage of selected sectors sometimes even varies from country to country for the same indicator.[3] Therefore, any sectoral information used for the present taxonomy is converted to the ISIC revision 4 sectoral classification to follow the 38-category breakdown proposed in the System of National Accounts (SNA). In some cases, where information is only available at a higher level of aggregation for a few sectors, the same value of the more aggregated sector is attributed to all underlying subsectors.[4] Given the availability of detailed sectoral information across different data sources and coverage considerations, the indicators of AI intensity (including the summary indicator) focus on manufacturing (excluding coke & petroleum), construction and business services.[5] The detailed list of sectors is reported in Table A A.1 in Annex A. In order to provide a more comprehensive picture, any sectoral information available is reported in Annexes, discussing further key insights in the main text.

Geographical coverage: Some indicators are only available for a subset of countries. In a related analysis focusing on digital intensity of sectors, Calvino et al. (2018[1]) addressed that by constructing a taxonomy based on the largest common denominator of countries for which all sector-specific information was present. Considering challenges in the availability of data to measure AI, we deviate from that approach trying to maximise the number of countries underlying each indicator, to the extent possible given data availability and computational constraints. Still, the extent to which the measures computed generalise to other countries, beyond those covered, remains an empirical question. The precise country coverage for each indicator is elaborated on in what follows.

Time coverage: The available time series vary significantly at the country-sector level across the different indicators. Considering this variation, the most recent years available, which are also most common across the different data sources, are prioritised. To reduce measurement error and avoid simply capturing time-specific features or shocks, the indicators are calculated as averages over multiannual windows, which differ in length depending on data availability. Furthermore, considering the specificities and availability of different data, the reference period may differ across indicators. For instance, there is at least an 18-months delay between when a patent application is filed and when it gets published in the different jurisdictions where it has been applied. Additionally, patent applications also need to be linked with sectoral information from Orbis©, which is usually only made available with a 2-year delay. As a result, capturing AI patents for the AI innovation indicator after 2021 is challenging. Conversely, online vacancy data can be collected timelier, allowing for the AI human capital indicator to be constructed across a more recent time period.

AI human capital

AI has the potential to significantly transform business operations, as AI-driven products and services become a fundamental aspect of many workers' activities, sometimes even unnoticed. Although there is no consensus in the existing literature about the overall employment implications of the adoption of AI technologies, for AI technologies to be developed and ultimately used, human input is needed (OECD, 2023[5]). Humans play a crucial role in understanding and managing data collection, storage, and usage within complex economic systems to make meaningful predictions. Their expertise is essential for developing technologies such as real-time language translation tools (Borgonovi, Hervé and Seitz, 2023[6]) or cancer detection systems (Bi et al., 2019[7]). Even large language models, such as ChatGPT©, rely on individuals and their skills to be developed, maintained and put to use in economic systems.

One type of occupation particularly often associated with the development of AI systems are data scientists because they use data mining, natural language processing and machine learning techniques to transform large data into meaningful information for intelligent systems. The following Box 2.1 provides more information on their employment patterns by sector, focusing on the United States.

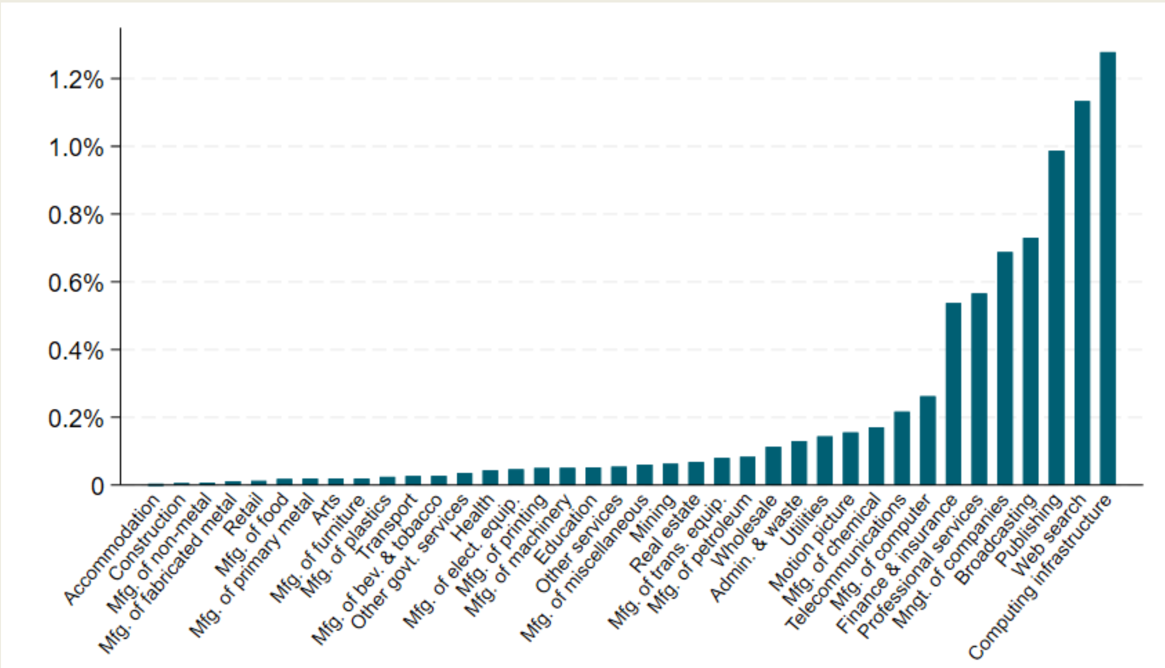
Box 2.1. Employment of Data Scientists in the United States

According to the U.S. Bureau of Labor Statistics, data scientists "develop and implement a set of techniques or analytics applications to transform raw data into meaningful information using data-oriented programming languages and visualization software [and] apply data mining, data modelling, natural language processing, and machine learning to extract and analyse information from large structured and unstructured datasets." These skills are critical for developing and maintaining AI systems, positioning data scientists as a key component of AI-related human capital.

As of May 2023, there were 192,710 data scientists employed in the United States, primarily in Professional Services, followed by the Finance and Insurance as well as Information sectors. However, when considering the share of data scientists relative to total employment within each industry, the Information sector ranks the highest. As shown in Figure 2.1, this is largely due to few industries within the sector that have a particularly high concentration of data scientists, namely 'Computing Infrastructure Providers, Data Processing, Web Hosting, and Related Services' (labelled Computing infrastructure), 'Web Search Portals, Libraries, Archives, and Other Information Services' (labelled Web search), 'Publishing' and 'Broadcasting'. In contrast, other industries within the Information sector, such as 'Telecommunications', employ comparatively fewer data scientists. Similar heterogeneity can be observed within Manufacturing, where the share of data scientists is twice as high for 'Computers' and three times as high for 'Chemicals' compared to 'Petroleum'.

Figure 2.1. Data Scientists, United States, May 2023

Percentage of employed Data Scientists, by NAICS industry



Note: The figure shows the percentage of data scientists, which is the total number of data scientists relative to all (excluding self-employed) workers in a given NAICS sector in May 2023. Data scientists are defined as SOC 15-2051 (hence excluding "Statisticians" (15-2041), "Cartographers and Photogrammetrists" (17-1021), and "Health Information Technologists and Medical Registrars" (29-9021)).
Source: U.S. Bureau of Labour Statistics.

To better understand the human capital embedded in the development and maintenance of AI, existing literature commonly relies on information provided in online vacancy texts (Samek, Squicciarini and Cammeraat, 2021[8]; Alekseeva et al., 2021[9]; Acemoglu et al., 2022[10]; Green and Lamby, 2023[11]; Manca, 2023[12]). Methodologies often leverage lists of AI-related keywords to classify AI-related vacancies which then provide insights into the skills demanded of the future workforce in the field of AI. Although exact empirical strategies vary in terms of the restrictiveness with which these keywords are applied, the share of AI-related vacancies remains small throughout the literature. A recent cross-country comparison of 14 OECD Member countries revealed a growing share of online vacancies requiring AI skills between 2019 and 2022, which never exceeded 1% of vacancies (Borgonovi et al., 2023[13]). However, these shares are very heterogeneous across industries, with professional activities, ICT, and manufacturing accounting for the majority of AI vacancies. Conversely, the share of postings requiring AI skills in Accommodation and food, Agriculture, and Transportation are very small.[6]

By relying on online vacancy data provided by Lightcast and drawing AI keywords from Baruffaldi et al. (2020[14]), we build on the work by Samek et al. (2021[8]) as well as Squicciarini and Nachtigall (2021[15]) to serve as a proxy for human capital demand for AI across sectors. The methodology requires vacancies to contain at least two AI-related skills belonging to different concepts or methodologies, only one of which may be a software-related skill. This empirical strategy is generally more conservative and reveals slightly lower shares of AI jobs compared to other studies.[7]

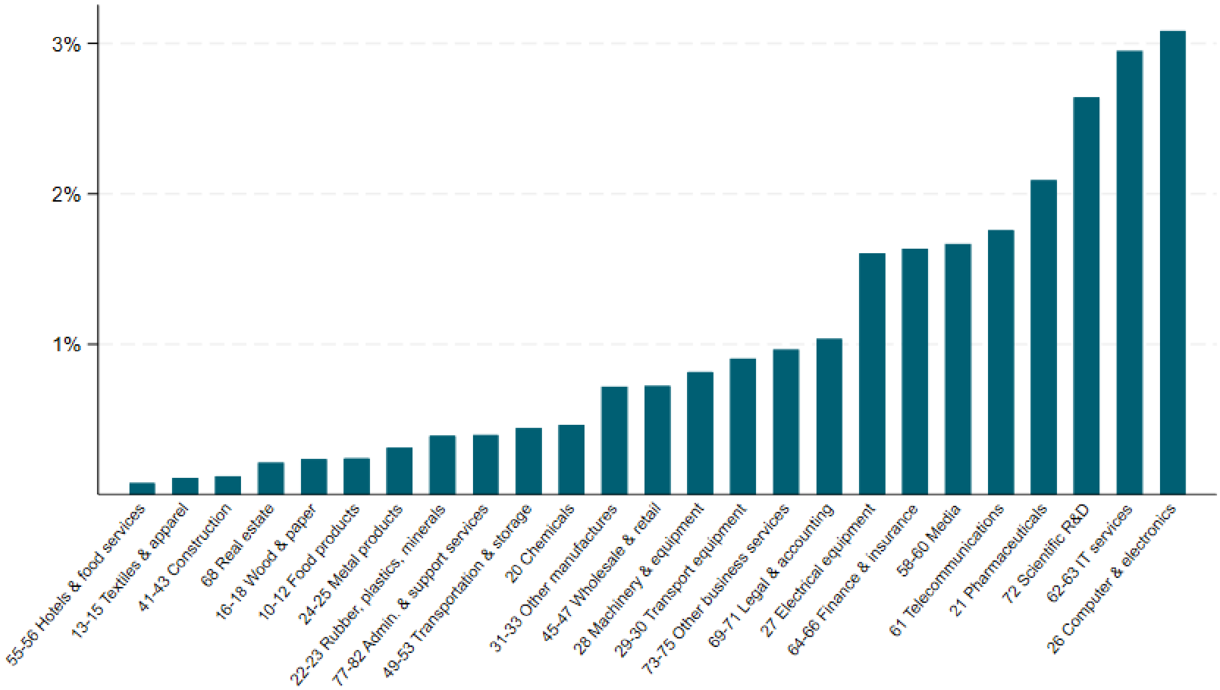
It should be noted that vacancies in general provide information on the demand for rather than the existing stock of human capital and it is not possible to disentangle whether such demand involves the replacement of existing labour, whether it aims at complementing the workforce, or even whether these vacancies are filled at all.[8] Moreover, given the online nature of the vacancies used, it is likely that growing firms and higher-skilled jobs, including AI-related jobs, are overrepresented, while low-skilled jobs are underrepresented, further exacerbating this bias (Cammeraat and Squicciarini, 2021[16]; Tsvetkova et al., 2024[17]; Vermeulen and Gutierrez Amaros, 2024[18]). After all, AI jobs are by their very nature part of the digital transformation and hence are expected to be posted online.

The AI intensity indicator related to the human capital component featuring in the final AI intensity taxonomy is shown in Figure 2.2. More specifically, it shows the number of AI-related online vacancies as a share of all online vacancies in the available data, averaged across the period 2018 to 2022 and across Canada, Singapore, the United Kingdom and the United States. Focusing on these four English-speaking countries enables the use of a consistent set of English keywords without the need to translate them into local languages, ensuring accuracy and uniformity when identifying AI-related vacancies.[9] In addition, it enables leveraging more detailed skill information, based on Lightcast's taxonomy of 32,000 unique competencies, knowledge, and specific software skills.[10]

Reaching around 3% of vacancies requiring AI skills, the top three most AI intensive sectors are manufacturing of 'Computer & electronics', 'IT services' and 'Scientific R&D'. These are sectors at the core of the digital transformation (see also Calvino et al., (2018[1])), it is therefore not unexpected that they score highest in their demand for AI-related human capital. These sectors are followed by other manufacturing sectors, such as 'Pharmaceuticals' and 'Electrical equipment', as well as other business services like 'Media', 'Telecommunications' and 'Finance & insurance'. Therefore, this group of sectors is positioned in the top half of the sectoral distribution of the AI human capital indicator.

Figure 2.2. Share of AI-related online vacancies by sector, 2018-22

Number of online vacancies containing AI skills as a share of all vacancies, Canada, Singapore, United Kingdom and United States

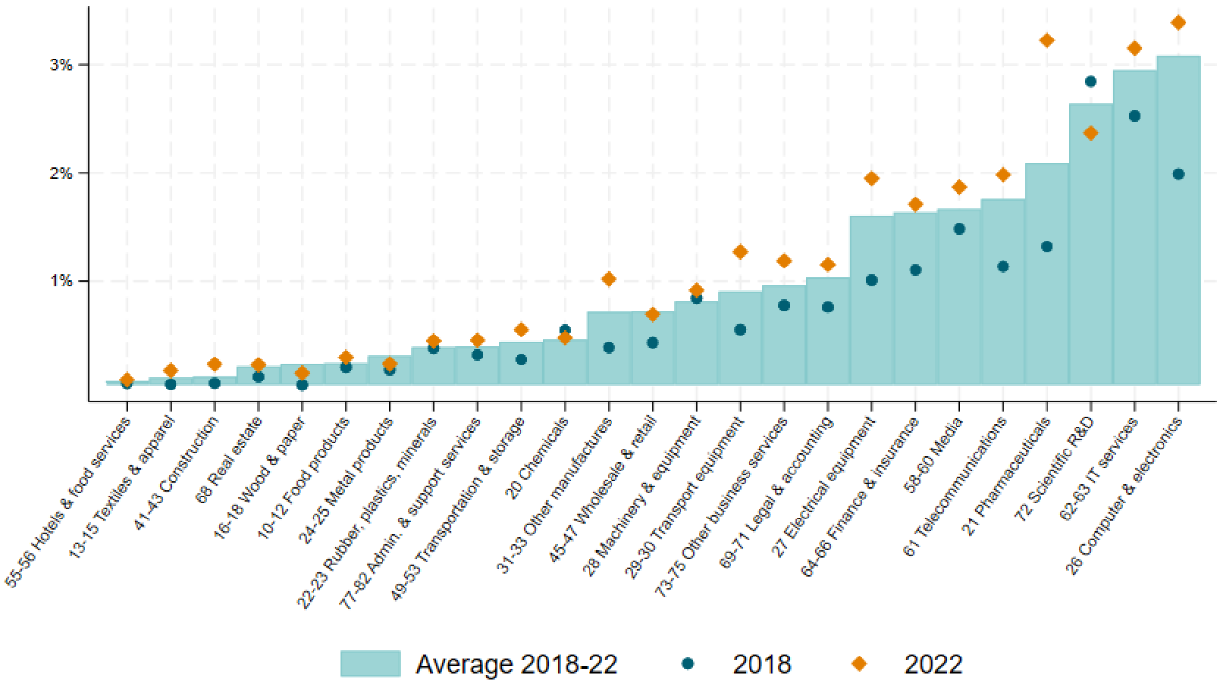


Note: This figure shows the percentage of online vacancies containing AI skills in a given sector, which is the number of AI-related vacancies relative to all vacancies in a specific sector. Only sectors for which the AI intensity taxonomy is constructed are displayed, i.e., manufacturing (excluding coke & petroleum), construction and business services. Sector-specific values are unweighted averages over the years 2018-22 across countries, namely: Canada, Singapore, United Kingdom and United States.
Source: authors' calculations based on Lightcast data.

While focusing on a four-year average reduces the impact of shorter-term shocks, exploring differences over time may provide additional relevant insights, possibly uncovering further specificities in sectoral patterns before and over COVID-19. A first feature that emerges in Figure 2.3, which displays the share of AI-related vacancies for 2018 and 2022, is the relative stability of rankings in the two periods, especially when focusing on sectors that are covered in the summary indicator. Correlating the quartile rankings of sectors for each year or each country lends further support that the indicator reflects, at least to a good extent, sectoral rather than time- or country-specific characteristics.[11]

Figure 2.3. Share of AI-related online vacancies by sector, 2018 versus 2022

Number of online vacancies containing AI skills as a share of all vacancies, Canada, Singapore, United Kingdom and United States



Note: This figure shows the percentage of online vacancies containing AI skills in a given sector and year, which is the number of AI-related vacancies relative to all vacancies in a specific sector and year. Only sectors for which the AI intensity taxonomy is constructed are displayed, i.e., manufacturing (excluding coke & petroleum), construction and business services. Sector-specific values are unweighted averages across countries, namely: Canada, Singapore, United Kingdom and United States. The bar shows the unweighted average across countries and the years 2018-22.
Source: authors' calculations based on Lightcast data.

Moreover, a significant increase in the share of AI vacancies across almost all sectors becomes evident. This should however be interpreted with some caution, also considering possible sector-specific differences in time coverage. Some of the largest increases are observed in sectors which are characterised by a relatively low demand for AI talent in 2018 and are also not covered in the summary indicator, such as 'Agriculture' and 'Public services' (see Figure A B.1 in Annex B). Among the sectors that feature in the summary indicator reported in Figure 3.1, 'Electrical equipment', 'Computer and Electronics' and 'Pharmaceuticals', which already had high shares of AI vacancies in 2018, exhibited significant increases in demand. Significant increases in the demand for AI-related skills in 'Pharmaceuticals' may reflect the expanding relevance and application of AI to different steps of development and deployment of pharmaceutical products, such as medicines or vaccines (World Health Organization, 2024[19]). Only very few sectors exhibited a decrease in demand for AI talent, namely 'Scientific R&D' and manufacturing of 'Chemicals'. However, these may not necessarily be indications of a persistent decreasing trend in the demand for AI-related human capital in these sectors as the share of these vacancies fluctuates and was highest for 'Scientific R&D' in 2021. It rather highlights the relevance of calculating each indicator over annual windows to reduce measurement error and avoid capturing time-specific features or shocks. This is of particular relevance since, as previously highlighted, the vacancy data captures the onset and thus the peak of the COVID-19 pandemic period.

AI innovation

Innovation, such as improving quality and upgrading processes, is a key motivation for U.S. firms to invest in advanced technologies, such as AI or robotics (Acemoglu et al., 2022[20]). Therefore, innovation is an important component when aiming to provide a comprehensive picture of the digital transformation, especially when focusing on a complex and multifaceted technology such as AI. With administrative micro-data on intellectual property rights constituting a rich source of information, patents can be leveraged to shed light on the developments in the field of AI. They describe what they intend to protect and contain information about technological scope, allowing specific technology domains to which a patented invention pertains (such as AI) to be identified.

Patents are a key measure of innovation output and are widely used in the academic and policy literature analysing the patterns and implications of technological change. In relation to AI, patents have been for instance used to better understand technological trajectories of AI inventions. Hötte et al. (2023[21]) compare different AI patent classification approaches to explore concentration patterns and other characteristics, while Dernis et al. (2021[22]), in addition to patents, also examine characteristics of AI-related trademarks. Martinelli et al. (2021[23]) use patents to examine if technologies, such as AI, that are ought to enable the 'factory of the future', display characteristics of General-Purpose Technologies. More recently, studies also examine specific sets of AI technologies. For instance, Santarelli, Staccioli and Vivarelli (2022[24]) map the knowledge base in robotics and AI technologies, distinguishing core and other related innovations, whereas Calvino et al. (2023[25]) use AI-related patents to provide a first exploratory analysis of technologies and applications that are at the core of recent advances in AI.[12]

In this study, shares of AI patents by sector are used as a proxy of AI innovation. AI-related patents are identified using the approach described in Baruffaldi et al. (2020[14]) which relies on a combination of AI keywords retrieved in the abstract or title of patent documents and on a selection of International Patent Classification (IPC) and Cooperative Patent Classification (CPC) codes that relate to AI. The list of AI keywords was derived from a text mining approach performed on scientific literature dedicated to AI. Both the relevant classification codes and keywords are listed in Annex C.[13]

However, by relying on a patent-based approach, this indicator may not comprehensively capture all AI innovations. Firstly, not all innovating firms choose to rely on patents to protect their inventions and the geographical distribution may depend on many factors, such as patenting costs, time for patent granting, or patent office rules (Khan and Dernis, 2006[26]). Secondly, not all AI technologies can be protected by patents. For instance, AI technologies, such as generative AI, are evolving at a fast pace and can impact the innovation process itself when used as a tool in the inventive process, which is not patentable (World Intellectual Property Organization, 2024[27]). Nevertheless, patents have long been recognised to represent a good proxy for inventive output (Griliches, 1990[28]). Notably, Dernis et al. (2021[22]) find AI actors to rely more heavily on patents to protect new AI inventions internationally than on trademarks to protect new AI-products.

The time frame under analysis focuses on families of patent applications filed between 2017 and 2021 within the Five Intellectual Property offices (IP5 patent families), namely the United States Patent and Trademark Office (USPTO), the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Patent Office (KIPO), and the China National Intellectual Property Administration (CNIPA), covering sets of patents issued and pre-grant patent publications.[14] Data derive from the Autumn 2023 version of the Worldwide Patent Statistical Database of the EPO, also known as the PATSTAT database, that is maintained by the OECD STI Micro-data Lab.

To retrieve sectoral information, each AI-related patent is combined with information on firms' financial accounts sourced from Moody's Orbis© database. Orbis© is a widely used commercial database that aggregates company accounts and financial reports across countries, covering almost 400 million companies and entities worldwide of which around 41 million contain detailed financial information on firm characteristics, including detailed sector of activity. The names of patent applicants were linked to company names provided in Orbis©, using series of string-matching algorithms (Squicciarini and Dernis, 2013[29]). For the purpose of this analysis, only countries with matching rates to Orbis© above 60% of their patent portfolio, and with more than 15 AI-related patents in half or more of the relevant sectors between 2017 and 2021 are considered. As a result, the following countries are included: Australia, Austria, Belgium, Denmark, Finland, France, Germany, Israel, Italy, Japan, Netherlands, Spain, Sweden, Switzerland, United Kingdom and the United States.

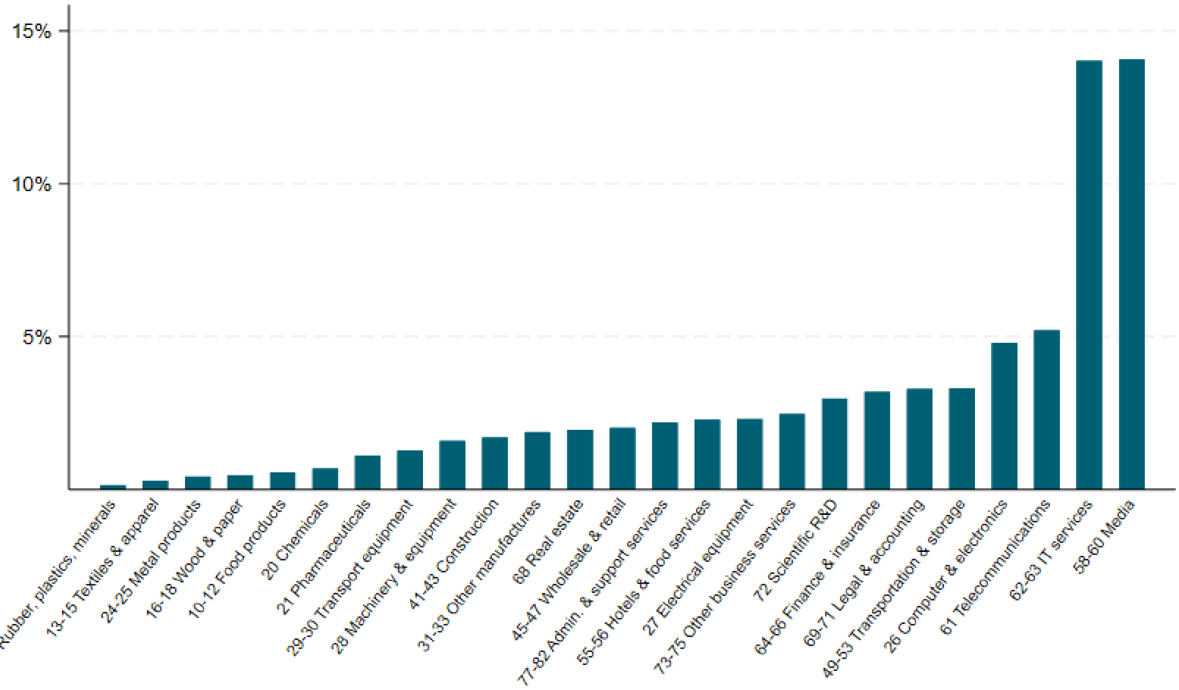
Using a longer time period for this indicator is important considering the significant increase in the number of patent families in AI and the acceleration in AI developments especially since 2015 (Dernis et al., 2021[22]). However, as previously mentioned, capturing AI patents for this indicator beyond 2021 is challenging, implying that the observation period precedes the more recent boom in generative AI. Firstly, there is at least an 18-months delay between the timing a patent is filed and when it gets published in the different jurisdictions it has been applied. Secondly, those patents then also need to be linked with sectoral information from Orbis©, which is usually only made available with a 2-year delay.

Keeping in mind these challenges, an examination of the cross-country within-sector average of the AI patenting activity, as presented in Figure 2.4, sheds light on the landscape of AI innovation. It becomes apparent that ‘IT services’ and ‘Media’ emerge as the sectors with highest shares of AI patents. Around 14% of patent applications filed between 2017 and 2021 are AI-related in the respective sector. This underscores how these sectors may be driving a significant proportion of AI innovations. These are indeed the sectors where some of the largest IT service providers tend to operate, with the ‘Media’ sector relevantly including the subsector identifying software publishing. This is followed, although with a much lower AI patenting activity of around 5% each, by ‘Telecommunications’ and the manufacturing of ‘Computer & electronics’, sectors that are key in the manufacturing of ICT equipment and in the provision of telecommunication services. Except for the latter, most sectors in manufacturing tend to show relatively lower shares of AI patents.

Therefore, some of the most AI intensive sectors are similar when measured in terms of human capital needed to develop AI and the AI-related innovation output produced. Manufacturing of ‘Computer & electronics’, ‘IT services’, ‘Telecommunications’ and ‘Media’ make up the top four sectors in terms of AI patenting activity and are within the top six sectors, together with ‘Scientific R&D’ and manufacturing of ‘Pharmaceuticals’ in terms of AI human capital.

Figure 2.4. Share of AI patents by sector, 2017-21

Number of AI patents as a share of total IP5 patents by sector, cross-country average

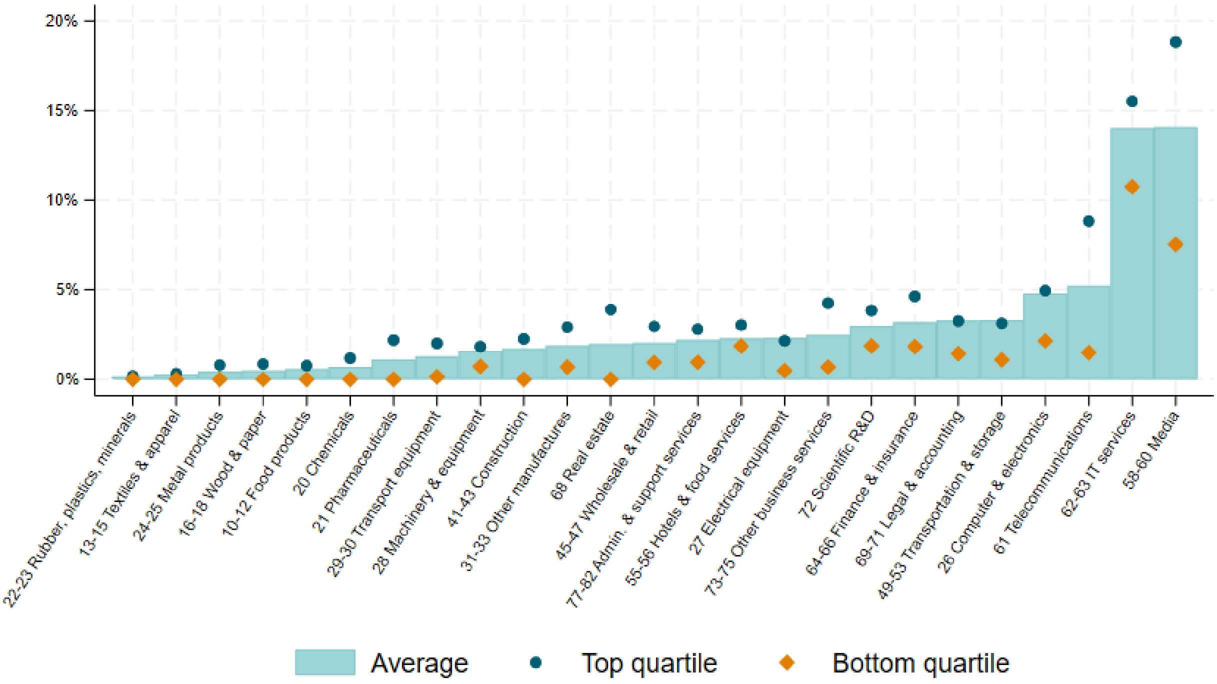


Note: This figure shows the percentage of AI patents in a given sector, which is the number of AI patents relative all IP5 patents in a specific sector. Only sectors for which the AI intensity taxonomy is constructed are displayed, i.e., manufacturing (excluding coke & petroleum), construction and business services. Sector-specific values are unweighted averages over the years 2017-21 and countries. Only countries with matching rates to Orbis above 60%, and with more than 15 AI patents in the period considered are included in the figure, namely: Australia, Austria, Belgium, Denmark, Finland, France, Germany, Israel, Italy, Japan, Netherlands, Spain, Sweden, Switzerland, United Kingdom and the United States.
Source: OECD, STI Micro-data Lab: Intellectual Property Database, http://oe.cd/ipstats, March 2024.

However, some degree of cross-country heterogeneity emerges, especially in sectors featuring high shares of AI patents. This is shown in Figure 2.5 and in Figure A C.1 in Annex C, where bars mirror the average share of AI patents presented in Figure 2.4, dots represent the country at the 75th percentile, and diamonds show the country at the 25th percentile of the distribution of AI patent shares. In fact, sectors exhibiting the more considerable degree of cross-country variation are the ‘Media’, ‘Telecommunication’ and ‘IT services’ sectors. However, their relative position in the ranking does not considerably change even when considering the median rather than the average rankings.[15]

Figure 2.5. Country dispersion of AI patents shares by sector, 2017-21

Number of AI patents as a share of total IP5 patents by sector



Note: The bars in the Figure shows the percentage of AI patents in a given sector, which is the number of AI patents relative all IP5 patents in a specific sector ("Average"). The dots ("75th percentile") represent the country at the 75th percentile of the distribution in the sector and period considered. The diamonds ("25th percentile") represent the country at the 25th percentile of the distribution in the sector and period considered. Only sectors for which the AI intensity taxonomy is constructed are displayed, i.e., manufacturing (excluding coke & petroleum), construction and business services. Sector-specific values are unweighted averages over the years 2017-21 and countries. Only countries with matching rates to Orbis above 60%, and with more than 15 AI patents in half or more of the relevant sectors in the period considered are included in the figure, namely: Australia, Austria, Belgium, Denmark, Finland, France, Germany, Israel, Italy, Japan, Netherlands, Spain, Sweden, Switzerland, United Kingdom and the United States.
Source: OECD, STI Micro-data Lab: Intellectual Property Database, http://oe.cd/ipstats, March 2024.

Barrier-adjusted AI exposure

Workers stand at the forefront of those who could either significantly benefit from or be adversely affected by the diffusion of AI. On the one hand, AI has the potential to enhance productivity, create new job opportunities, and augment worker capabilities. On the other hand, it also poses the risk of displacing workers, particularly in roles susceptible to task replacement. The recent literature has derived proxy measures to assess the degree to which specific occupations are likely to overlap with tasks that AI can perform, commonly referred to as "AI exposure" measures.

The term AI exposure generally reflects the potential for AI to be integrated into occupations, based on the tasks and skills that characterise each job. Webb (2019[30]) measures AI exposure by evaluating the similarity between AI patent applications and the tasks defining an occupation. Briggs and Kodnani (2023[31]) identify work activities exposed to AI and assess whether these activities, within an occupation, are sufficiently low in complexity for AI to perform them. Eloundou et al. (2023[32]), in the context of generative AI, define exposure as the extent to which access to large language models can reduce the time required for a human to perform a specific work activity or complete a task by at least 50 percent. Among these measures, the 'AI at the occupational level' (AIOE) index by Felten, Raj, and Seamans (2021[4]) is the most widely used.[16] It aims to capture the relative degree of overlap between key AI applications and the abilities required to perform a given occupation (see Box 2.2 for details on the methodology). Relatedly, Felten, Raj, and Seamans (2023[33]) update the AIOE score to account for advances in Large Language Modelling applications, like ChatGPT©.[17]

Extensions to AI exposure measures have aimed to account for the mediating factors influencing the exposure of workers to AI. Indeed, the measure is usually agnostic as to the effects of AI on the occupation, which could involve substitution or augmentation depending on various factors associated with the occupation itself and the broader sectoral characteristics (Felten, Raj and Seamans, 2021[4]). For instance, Gmyrek et al. (2023[34]) categorise occupations based on their potential for automation or augmentation, depending on the distribution of AI-automation scores for the tasks within each occupation. The framework by Pizzinelli et al. (2023[35]) further develops a complementarity measure incorporating social, legal, and technical factors related to occupations, aimed at capturing the potential of exposure to AI to translate in productivity gains for workers or risks of displacement. More recently, the work by Svanberg et al. (2024[36]) looks at the technical feasibility and economic attractiveness of implementing AI systems, since usual exposure measures just focus on the potential for AI to affect occupations.[18]

More generally, sectoral characteristics and constraints to AI use, including potential barriers to adoption, may be significant factors influencing the actual exposure of workers to AI. Recent data at the sectoral level from the Business Trends and Outlook Survey (BTOS), U.S. Census Bureau (Bonney et al., 2024[37]) highlight that sectors tend to experience different barriers when it comes to the use of AI. Notably, Figure 2.6 reports the relative ranking positions of U.S. sectors according to three key barriers in the use of AI, namely “Costs and Maturity”, “Operational and Skills”, and “Regulatory and Ethical” barriers. These highlight the potential challenges that firms across sectors may face in terms of use of AI technologies. The Costs and Maturity barrier captures the potential internal constraints to AI use related to financial expenses and the developmental stage of AI technology. The Operational and Skills barrier highlights limitations in AI use related to insufficient knowledge about AI capabilities, a shortage of skilled personnel, inadequate data, and prior unsatisfactory experiences with AI applications. The Regulatory and Ethical barrier reflects the potential external constraints imposed by privacy and security concerns, laws and regulations, as well as concerns related to AI bias (see Box 2.3 and Annex D for further information). The relative rankings for these barriers offer some policy-relevant insights into the specific obstacles firms may encounter in integrating AI into their operations.

As shown in panel (A) of Figure 2.6, “Costs and Maturity” barriers appear highest in ‘Professional and Scientific’, ‘Information’ and ‘Finance and Insurance’ sectors. These may be due to the high costs associated with developing and implementing highly specialised AI applications in these fields, as well as the rapid changes in the industry requiring constant updates and investments. For example, cutting-edge AI technologies may not yet be fully developed or tested in Professional and Scientific fields, making use riskier and less certain. This appears aligned with the relatively high ranking of the ‘Health Care’ sector, where implementing AI usually requires substantial investment in infrastructure and training. Panel (B) of Figure 2.6 shows that ‘Professional and Scientific’, ‘Finance and Insurance’ and ‘Health Care’ rank highest also in terms of “Regulatory and Ethical” barriers. Indeed, firms in these sectors usually handle highly sensitive data, which likely require higher privacy and security measures that may affect AI use decisions.

Box 2.2. The AI exposure measure (AIOE) by Felten, Raj, and Seamans (2021)

Felten, Raj, and Seamans (2021[4]) derive a measure of exposure to AI at the occupational level (AIOE), focusing on data from the United States, capturing the degree of overlap between AI applications and the abilities needed for occupations. The AIOE index is computed by connecting 10 AI applications (from the Electronic Frontier Foundation, EFF) with 52 occupational abilities (from O*NET). As a first step, the authors derive a measure of relatedness for each combination of AI application and occupational ability, bounded between 0 and 1. This measure comes from a crowd-sourced data set, constructed using survey responses of "gig workers" from Amazon's mTurk web service. The authors organise information into a matrix that connects the 10 AI applications to the 52 O*NET occupational abilities, where each combination has a score x_{ij}, where i indexes the AI application and j indexes the occupational ability. Using the measures of application-ability relatedness from this matrix, the authors calculate an ability-level exposure A_{ij} as a sum of the scores x_{ij}:

$$A_{ij} = \sum_{i=1}^{10} x_{ij}.$$

In a second step, the authors weight the ability-level AI exposure A_{ij} by the ability's prevalence (L_{ij}) and importance (I_{ij}) within each occupation k, as measured by O*NET. Notably, the authors multiply the ability-level AI exposure by the prevalence and importance scores for that ability within each occupation, scaled so that they are equally weighted. Felten, Raj, and Seamans (2021[4]) thus obtain the following occupation-level AIOE measure:

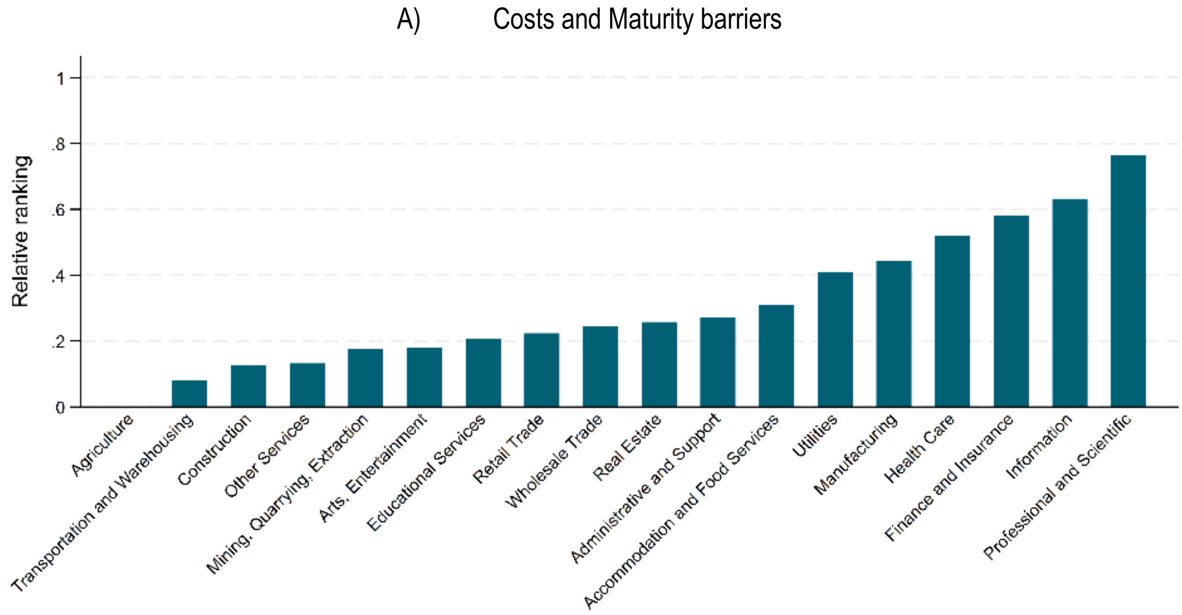
$$AIOE_k = \frac{\sum_{j=1}^{52} A_{ij} \cdot L_{jk} \cdot I_{jk}}{\sum_{j=1}^{52} L_{jk} \cdot I_{jk}}.$$

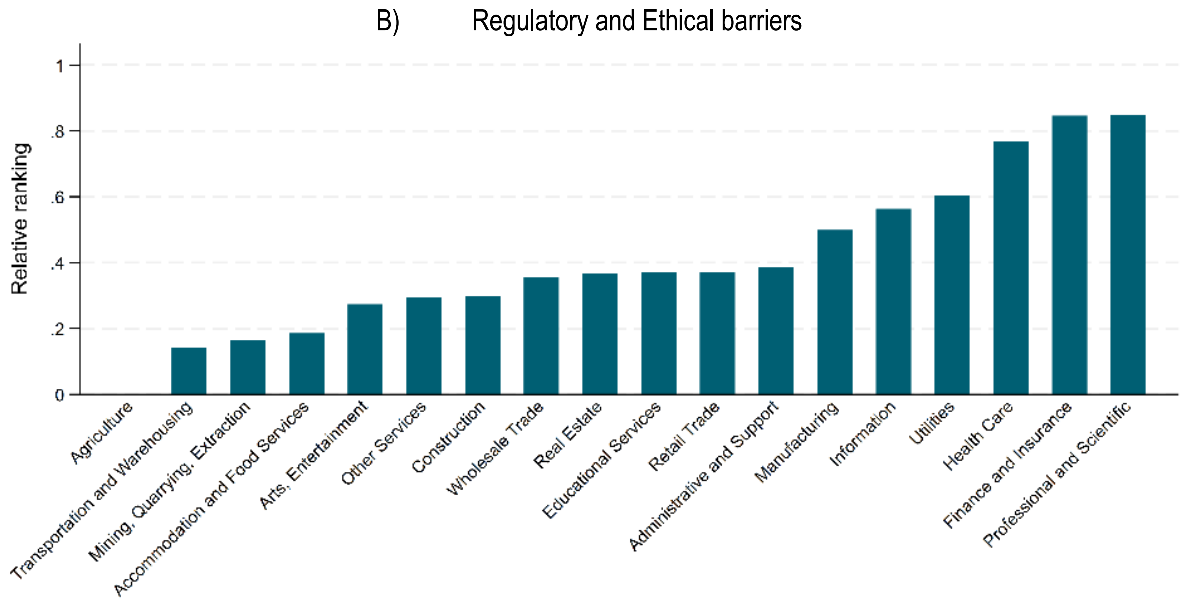
The authors collapse the occupations at the eight-digit to the six-digit Standard Occupational Classification (SOC) level by calculating the mean AIOE within the six-digit SOC level. They standardise the AIOE measures such that the mean across occupations is zero and the standard deviation is one. The scaling provides a measure of the relative exposure to AI across occupations. Lastly, the authors construct the industry level measure of AI exposure (AIIE) by aggregating the AIOE across occupations within an industry. In particular, the authors construct the AIIE measure by taking a weighted average of the AIOE using industry employment in the United States based on the four-digit North American Industry Classification System (NAICS) in 2019. This measure allows to track which industries have the highest exposure to AI.

Note: The ten applications are: "abstract strategy games", "real-time video games", "image recognition", "visual question answering", "image generation", "reading comprehension", "language modelling", "translation", and "speech recognition". In the procedure, the authors equally weight each application and do not consider potential interactions across applications. The O*NET database defines and describes professions in the workplace in the United States. It uses 52 distinct abilities to characterise the workplace activities of each occupation.

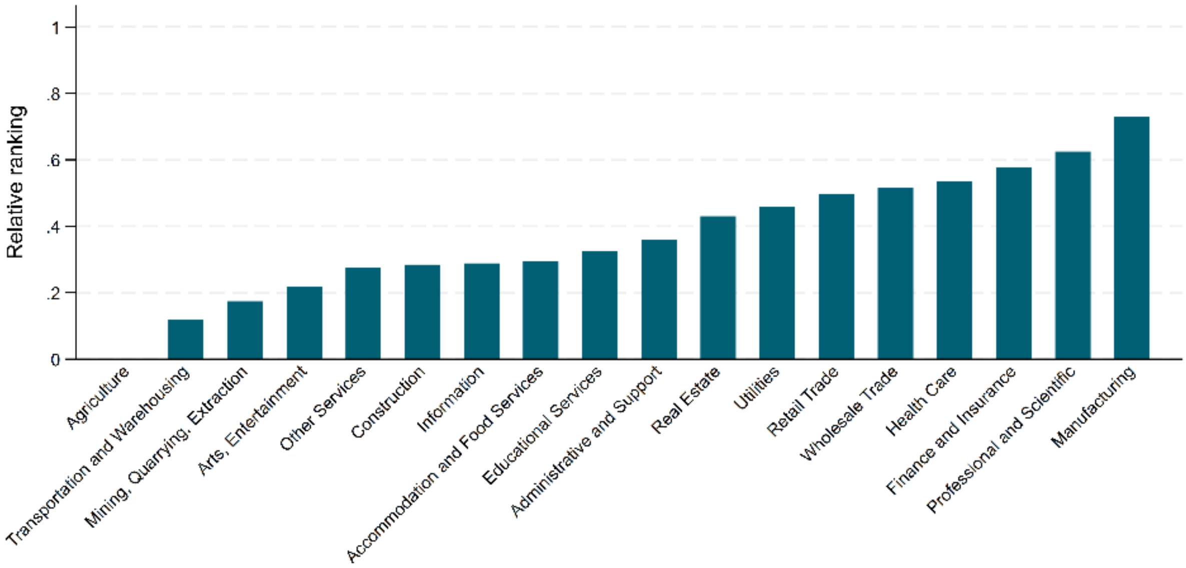
Lastly, panel (C) of Figure 2.6 focuses on "Operational and Skills barriers", revealing that 'Manufacturing' and 'Professional and Scientific' sectors rank the highest in this regard. This likely suggests that these sectors experience difficulties in using AI due to a lack of skilled workforce, inadequate data, or other limitations in the use of AI capabilities. The high ranking for the Manufacturing sector suggests that despite the potential benefits of AI in automating and optimising production processes, there may be notable gaps in transposing AI tools in production processes, likely due to the need of specialised training.

Figure 2.6. Sectoral barriers to AI use





C) Operational and Skills barriers



Note: The figure shows the relative ranking of NAICS 2-digit sectors according to three AI use barriers, namely "Costs and Maturity", "Operational and Skills", and "Regulatory and Ethical" barriers. NAICS 2-digit sectors are Agriculture (11 Agriculture, Forestry, Fishing and Hunting); Mining, Quarrying, Extraction (21 Mining, Quarrying, and Oil and Gas Extraction); Utilities (22); Construction (23); Manufacturing (31–33); Wholesale Trade (42); Retail Trade (44–45); Transportation and Warehousing (48–49); Information (51); Finance and Insurance (52); Real Estate (53 Real Estate and Rental and Leasing); Professional and Scientific (54 Professional, Scientific, and Technical Services); Administrative and Support (56 Administrative and Support and Waste Management and Remediation Services); Educational Services (61); Health Care (62 Health Care and Social Assistance); Arts, Entertainment (71 Arts, Entertainment, and Recreation); Accommodation and Food (72 Accommodation and Food Services); Other Services (81 Other Services—except Public Administration). Additional details on the ranking computation can be found in Annex D.
Source: authors' calculations based on BTOS data, U.S. Census Bureau.

The presence of barriers in the use of AI has the potential to affect diffusion patterns across firms and sectors, and also transposes to different levels of actual exposure, that may be more closely related to the actual potential of AI. Leveraging information from the BTOS, this work adjusts the AIIE[19] indicator by Felten, Raj, and Seamans (2021[4]) taking into account the role of barriers to the use of AI. Notably, the methodology rescales the relative AIIE measure and derives a barrier-adjusted AI exposure indicator (BAIIE) at the sectoral level.[20] The main rationale of the indicator is to account for the varying degrees of barriers that different sectors face in adopting and implementing AI technologies, thereby providing a representation of AI exposure that possibly reflects these constraints (see Annex D for further information on the methodology).

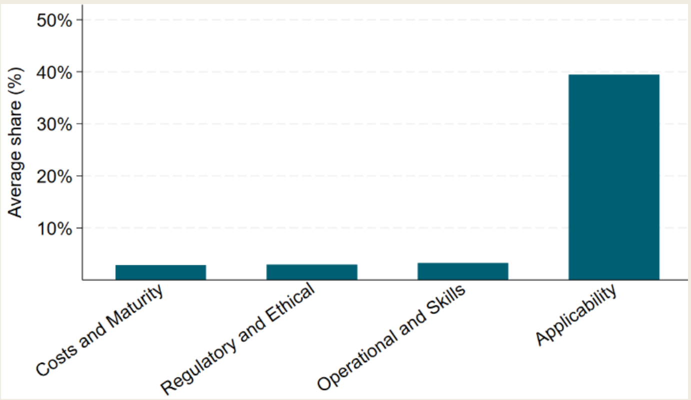
Box 2.3. Barriers to the use of AI

The Business Trends and Outlook Survey (BTOS) managed by the U.S. Census Bureau aims at capturing high-frequency changes in economic conditions through a qualitative survey representative of U.S. employer businesses. Supplemental content was added to the BTOS from December 2023 to February 2024 to provide detailed information about businesses' use of AI, e.g., in the last six months and six months into the future. Notably, the supplement asks firms that do not expect to use AI in the future why that might be the case, to uncover some of the key barriers to AI adoption or use (see Annex D for details about the questions and Bonney et al. (2024[37]) for extensive discussion).

Leveraging this data source, Figure 2.7 shows the average share of firms across sectors expecting not to use AI in the future, according to four main barriers. These are "Applicability", "Costs and Maturity", "Operational and Skills", and "Regulatory and Ethical" barriers, as derived from the aggregation of information from the BTOS (see also Annex D). The most significant barrier, accounting for over 40% of the share, is related to "Applicability", indicating that the most substantial portion of businesses find AI not applicable to their operations (Bonney et al., 2024[37]). "Operational and Skills" challenges, "Regulatory and Ethical" concerns, as well as "Costs and Maturity" show relatively lower average shares. As the AI exposure concept is inherently linked to the potential applicability of AI, this work focuses on the remaining barriers. Notably, "Costs and Maturity", "Operational and Skills", and "Regulatory and Ethical" barriers may be important impeding factors to the use of AI, especially where applicability barriers are less of a concern (see Annex D for further details). Relatedly, the relative importance of these barriers may affect the actual exposure of workers to AI.

Figure 2.7. AI use barriers across sectors

Share of firms reporting AI barriers across NAICS 2 sectors, by category



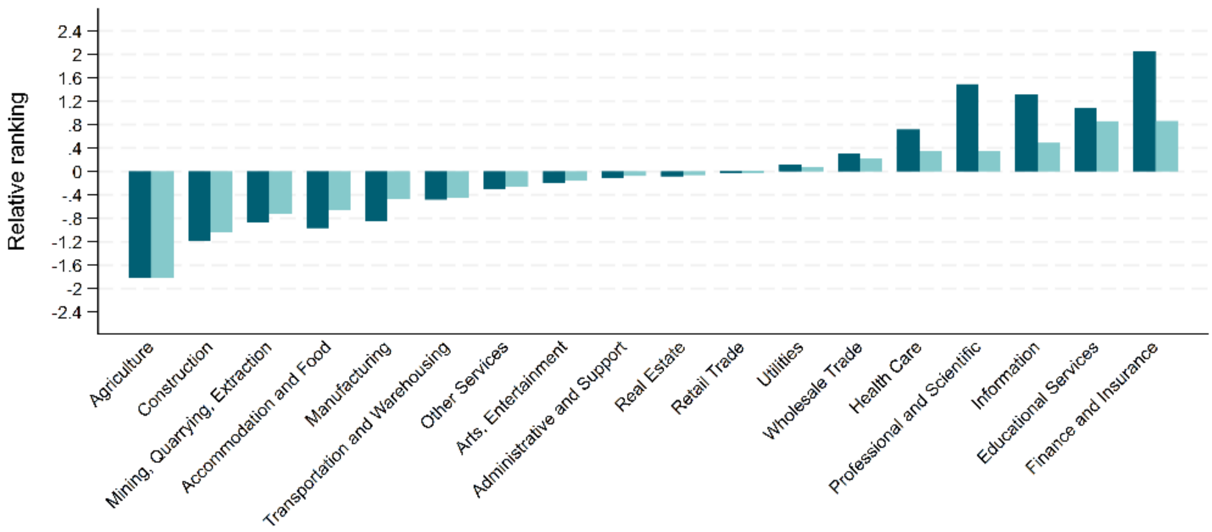
Note: The figure shows the average share of firms across NAICS 2-digit sectors that do not expect to use AI according to four barrier categories, namely "Applicability", "Costs and Maturity", "Operational and Skills", and "Regulatory and Ethical". Categories are derived from the aggregation of options in Question 37 of the BTOS survey. The Applicability barrier includes the option "AI is not applicable to this business". The Costs and Maturity barrier includes the options "Too expensive" and "AI is not a mature enough technology yet". The Operational and Skills barrier includes the options "Lack of knowledge on the capabilities of AI", "Lack of skilled workforce", "Lack of required data", and "Previous or current use of AI did not meet expectations". The Regulatory and Ethical barrier includes the options "Concerns about privacy/security", "Concerns about bias", and "Laws and regulations prevent or restrict use of AI". The "Other" category is not reported. Options are not mutually exclusive. To avoid double counting of firms within categories, for each barrier category the maximum share value across included options is used. Additional information is available in Annex D.
Source: authors' calculations based on BTOS data, U.S. Census Bureau.

Figure 2.8 reports the AIIE index by Felten, Raj, and Seamans (2021[4]) at the NAICS 2-digit level, and the corresponding BAIIE adjustments. In particular, each sectoral value of the AIIE index is downscaled according to the relative importance of a given barrier for that sector (as captured by the barriers indicator for each category; see again Figure 2.6).[21] Panel (A) focuses on AI exposure adjusted by Costs and Maturity barriers. Among others, it shows that 'Finance and Insurance', 'Information', and 'Professional and Scientific' sectors have lower barrier-adjusted AI exposure indicators (BAIIE) compared to their initial AI exposure indicators (AIIE). Importantly, the relative ranking across sectors appears to change, with 'Professional and Scientific' and 'Information' sectors resulting less exposed to AI in relative terms. Indeed, the high costs and developmental challenges associated with AI technologies in these sectors may result in a lower potential for workers to be exposed to AI. Conversely, 'Educational Services' result second in the BAIIE ranking. Indeed, as costs and maturity barriers appear lower in this sector vis-à-vis other sectors, actual AI exposure may result higher.

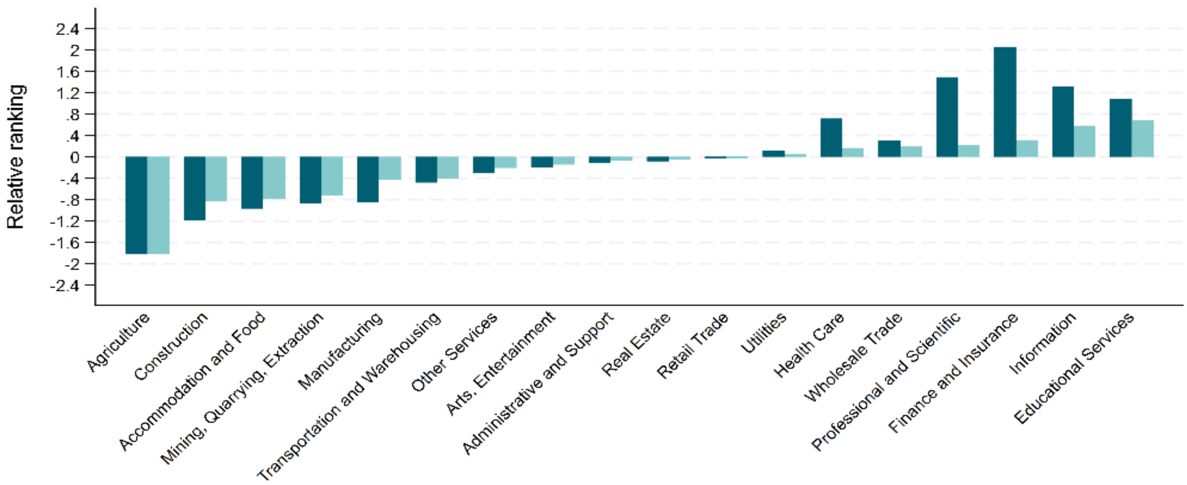
In a similar vein, Panel (B) focuses on the Regulatory and Ethical barriers adjustment. Also in this case, the relative ranking across sectors changes, with 'Professional and Scientific' and 'Finance and Insurance' sectors becoming less exposed to AI in relative terms. Conversely, the 'Educational Services' sector shows a higher barrier-adjusted AI exposure indicator (BAIIE) compared to the initial AIIE. This suggests that actual AI exposure may result higher in that sector, potentially due to lower sensitivity of the data involved and fewer regulatory constraints compared to finance, information, or scientific sectors. This results in a higher potential for AI exposure when these barriers are accounted for.

Figure 2.8. Relative ranking of sectors according to AIIE and BAIIE measures, by type of barrier

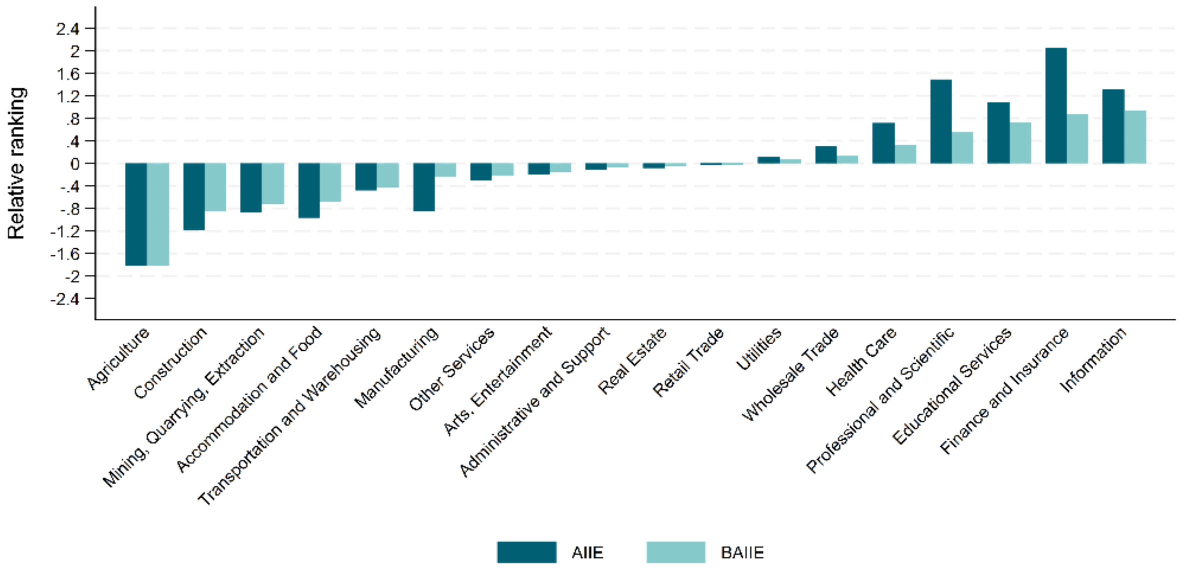
A) AI exposure adjusted by Costs and Maturity barriers



B) AI exposure adjusted by Regulatory and Ethical barriers



C) AI exposure adjusted by Operational and Skills barriers



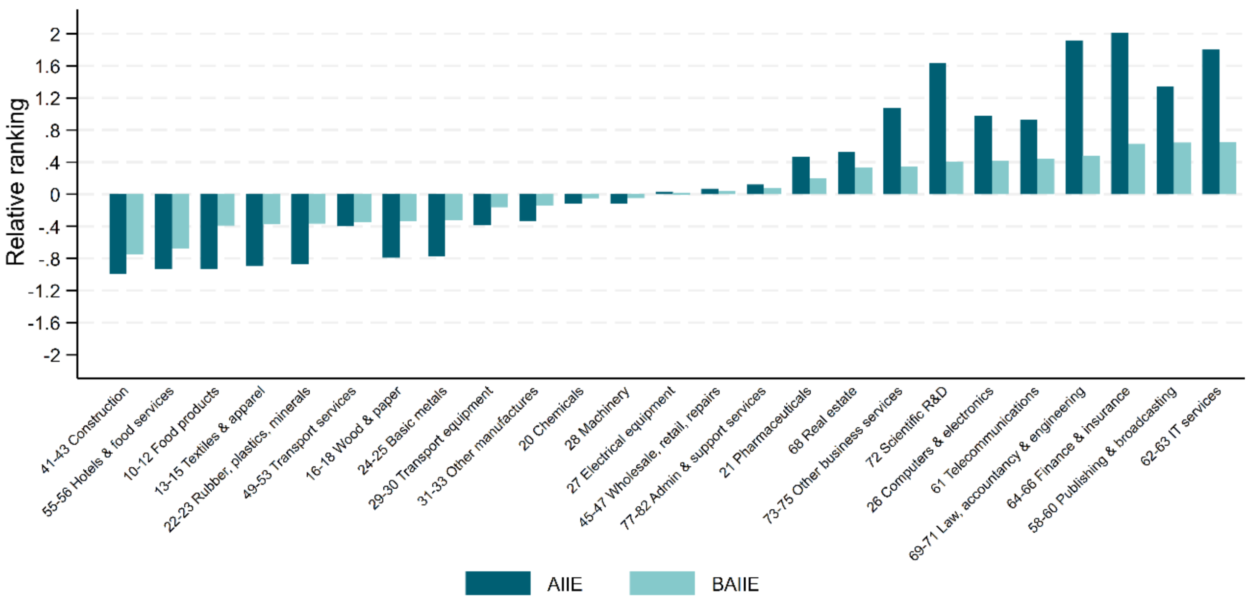
Note: The figure reports the AIIE and BAIIE measures at the NAICS 2-digit sectoral level. BAIIE measures are adjusted according to three barriers, "Costs and Maturity" (A), "Operational and Skills" (B), and "Regulatory and Ethical" (C). NAICS 2-digit sectors are Agriculture (11 Agriculture, Forestry, Fishing and Hunting); Mining, Quarrying, Extraction (21 Mining, Quarrying, and Oil and Gas Extraction); Utilities (22); Construction (23); Manufacturing (31–33); Wholesale Trade (42); Retail Trade (44–45); Transportation and Warehousing (48–49); Information (51); Finance and Insurance (52); Real Estate (53 Real Estate and Rental and Leasing); Professional and Scientific (54 Professional, Scientific, and Technical Services); Administrative and Support (56 Administrative and Support and Waste Management and Remediation Services); Educational Services (61); Health Care (62 Health Care and Social Assistance); Arts, Entertainment (71 Arts, Entertainment, and Recreation); Accommodation and Food (72 Accommodation and Food Services); Other Services (81 Other Services—except Public Administration). Bars are organised in increasing BAIIE order. Additional details on the BAIIE computation can be found in Annex D.
Source: authors' calculations based on BTOS data (U.S. Census Bureau) and Felten, Raj, and Seamans (2021[4]).

Lastly, Panel (C) focuses on Operational and Skills barriers. In this case, the 'Information' sector results the one with the highest ranking, indicating that exposure in this sector may be actually higher vis-à-vis other sectors, considering the lower relative barriers faced. Also in this case, the most notable change concerns 'Educational Services'. This sector moves up significantly as it likely faces fewer operational and skills barriers compared to other sectors. Some of the insights provided by Figure 2.8 appear particularly relevant from a policy perspective, as they may provide an approximation of the extent to which different barriers may constrain the potential of AI, supporting evidence-based policy making in this domain.

As the analysed barriers are not mutually exclusive, Figure 2.9 shows a composite adjusted indicator taking into account these three potential impediments to AI use, likely affecting the actual degree of AI exposure of sectors. The Figure reports the results for the AIIE and BAIIE measures at the SNA A38 level, consistently with other indicators of AI intensity, for the sectors included in the taxonomy in Section 3 (see Annex D for further information on the weighting and sectoral conversion). After taking into account the three barriers, 'IT Services' remains among the highest-ranked sectors in both AIIE and BAIIE, indicating a relatively stable AI exposure despite the barriers. 'Finance & Insurance', initially ranked very high in terms of AIIE, shows a significant shift downwards in the BAIIE ranking. Similarly, the 'Other business services' sector experiences a noticeable decrease in ranking from AIIE to BAIIE, reflecting the impact of barriers on its AI adoption potential. Conversely, 'Publishing & broadcasting' sector moves up in the rankings when adjusted for barriers, suggesting it is less affected by these constraints compared to its initial AIIE ranking.[22]

Figure 2.9. Barrier-adjusted AI exposure

Relative ranking of SNA A38 sectors according to AIIE and BAIIE measures



Note: The figure reports the AIIE and BAIIE measures at the SNA A38 sectoral level. Only sectors for which the AI intensity taxonomy is constructed are displayed, i.e., manufacturing (excluding coke & petroleum), construction and business services. The BAIIE measure adjusts the AIIE index taking into account the role of barriers "Costs and Maturity", "Operational and Skills", and "Regulatory and Ethical". Bars are organised in increasing BAIIE order. Additional details on the BAIIE computation can be found in Annex D. Figure A D.4. in Annex D reports the complete sectoral distribution.
Source: authors' calculations based on BTOS data (U.S. Census Bureau) and Felten, Raj, and Seamans (2021[4]).

AI use

Vacancy data to proxy demand for human capital or patents as a proxy of innovation may not necessarily capture the extent to which firms use AI technologies. Furthermore, as previously shown, the potential of AI depends not only on the degree to which specific occupations are likely to overlap with tasks that AI can perform but also on barriers, that may be specific to certain sectors, and that may constrain such potential. Beyond such constrains, technological diffusion does not occur instantaneously, but involves path dependencies, sequentiality and complementarities with other factors internal or external to firms, that may vary from sector to sector. In this context, an essential facet to characterise the extent to which AI relates to the activity of economic sectors is the actual use of AI.

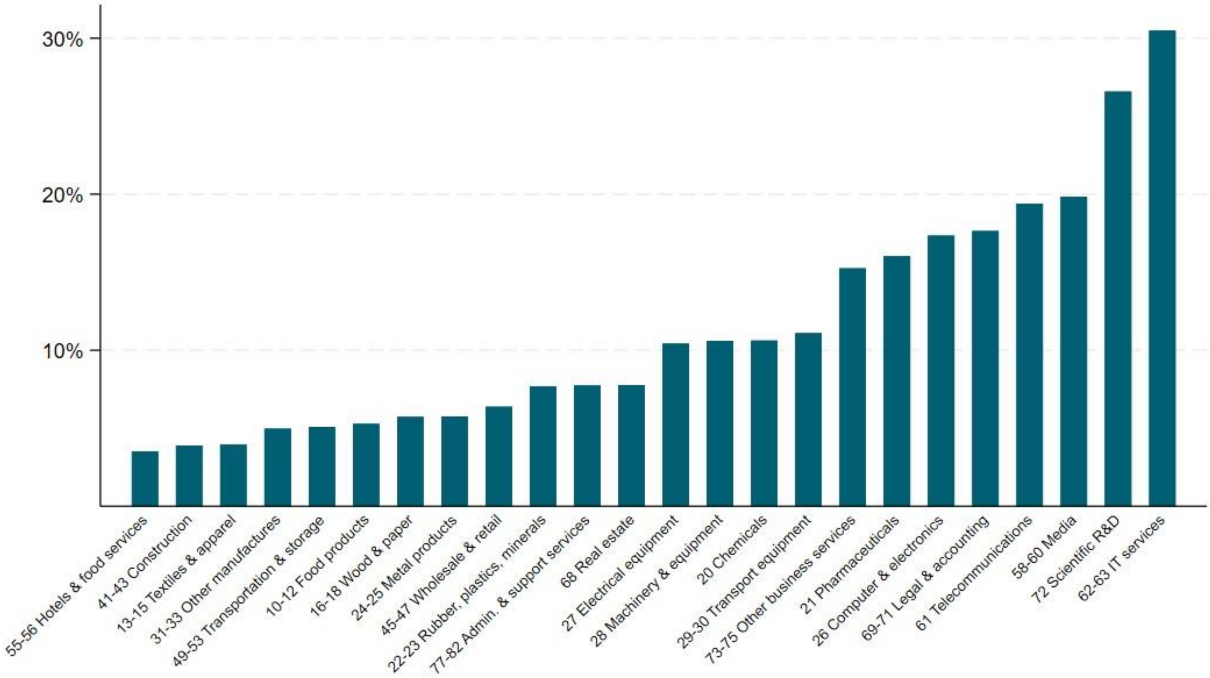
Cross-country evidence on the extent to which AI is used by firms is still limited, although recently emerging. Calvino and Fontanelli (2023[38]), for instance, recently used a common statistical code developed by the OECD in a distributed micro-data approach, which is executed in a decentralised manner on official firm-level surveys to better understand AI diffusion in firms across 11 countries. They find shares of AI adoption to be consistently higher in ICT and Professional services, with complementary assets, like skills and digital capabilities, positively linked to AI use. Using the Annual Business Survey in the United States, Zolas et al. (2020[39]) find less than 7% of firms to have used AI technologies in 2018. Distinguishing different technology types, McElheran (2024[40]) find that share to be even lower with firms in Manufacturing and Information having led in early AI adoption. Based on the BTOS data (September 2023-February 2024) discussed above, Bonney et al. (2024[37]) highlight an expected AI use rate of about 6.6 %. Similar diffusion patterns are evident in Germany, where less than 6% of firms were identified as AI users in the German 2018 Community Innovation Survey, with a high concentration in ICT and consulting services (Rammer, Fernández and Czarnitzki, 2022[41]). According to the Survey for Business Activities from Statistics, about 9% of Korean firms use AI (Cho et al., 2023[42]) in 2017.The latest OECD Digital Economy Outlook 2024 (OECD, 2024[43]) suggests that average AI adoption rates of firms with 10 employees or more across OECD Member countries are at 8%, with most OECD Member countries exhibiting shares of AI use lower than 10%.

The AI use indicator presented below leverages the latest publicly available data that are collected by National Statistical Institutes based on official surveys.[23] In particular, the indicator leverages the annual Eurostat survey on usage of ICT and e-commerce in enterprises, which contains information for EU27 countries, Norway and Türkiye. More specifically, firms with at least 10 persons engaged are considered to be AI users if they adopt any of the following seven technologies (see further details in Annex E): technologies performing analysis of written language (text mining); technologies converting spoken language into machine-readable format (speech recognition); technologies generating written or spoken language (natural language generation); technologies identifying objects or persons based on images (image recognition, image processing); machine learning (e.g. deep learning) for data analysis; technologies automating different workflows or assisting in decision making (AI based software robotic process automation); or technologies enabling physical movement of machines via autonomous decisions based on observation of surroundings (autonomous robots, self-driving vehicles, and autonomous drones).[24] The number of firms using AI as a share of all firms are broken down by NACE Rev. 2 categories (excluding agriculture, mining, public services, and finance as well as insurance) and for the purpose of this analysis averaged across the two most recent years available, namely 2021 and 2023.[25]

As shown in Figure 2.10, while shares are significantly higher, the composition of the most AI intensive sectors is rather similar when measured in terms of AI use, and AI human capital (Figure 2.2) or AI innovation (Figure 2.5). This is also the case when comparing the rankings of sectors with the BAIIE measure, which does not rely on shares (Figure 2.9), as further discussed in the next section. AI use is highest in ‘IT services’, with more than 30% of firms indicating to use at least one AI technology. AI intensity is second highest in ‘Scientific R&D’, which ranked also high in terms of AI human capital but lower in terms of AI innovation. ‘Media’ and ‘Telecommunications’ are also very AI intensive with almost one in five firms using AI, on average over the observed period.

Figure 2.10. Share of firms using AI by sector, 2021 and 2023

Number of firms using AI technologies as a share of all firms with at least 10 employees, EU27, Norway and Türkiye



Note: This figure shows the percentage of firms using AI in a given sector, which is the number of firms with at least 10 employees using any AI technology relative to all firms of the same size in a specific sector. Only sectors for which the AI intensity taxonomy is constructed are displayed, i.e., manufacturing (excluding coke & petroleum), construction and business services. Sector-specific values are weighted averages over the years 2021 and 2023 and EU27 (Austria, Belgium, Bulgaria, Croatia, Cyprus, Czechia, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Poland, Portugal, Romania, Slovak Republic, Slovenia, Spain, and Sweden), Norway and Türkiye, where the aggregate of EU27 was given a weight of 27/29, and Norway and Türkiye 1/29 each.
Source: Eurostat, https://doi.org/10.2908/ISOC_EB_AIN2

These observations are also confirmed when extending the sample for the AI use indicator to all OECD Member countries as shown in Figure A E.1 in Annex E. Although data is only available at the one-digit sector level, it is evident that the highest share of firms using AI is in the ‘Information and Communication’ (J) sector, which encompasses ‘IT services’ and ‘Media’. This is followed by ‘Professional, scientific & technical activities’ (M), which consists of ‘Scientific R&D’, ‘Legal & Accounting’ and ‘Other business services’ – all sectors which individually exhibit very high shares of AI use in Figure 2.10. This is also in line with the findings by Calvino and Fontanelli (2023[38]).

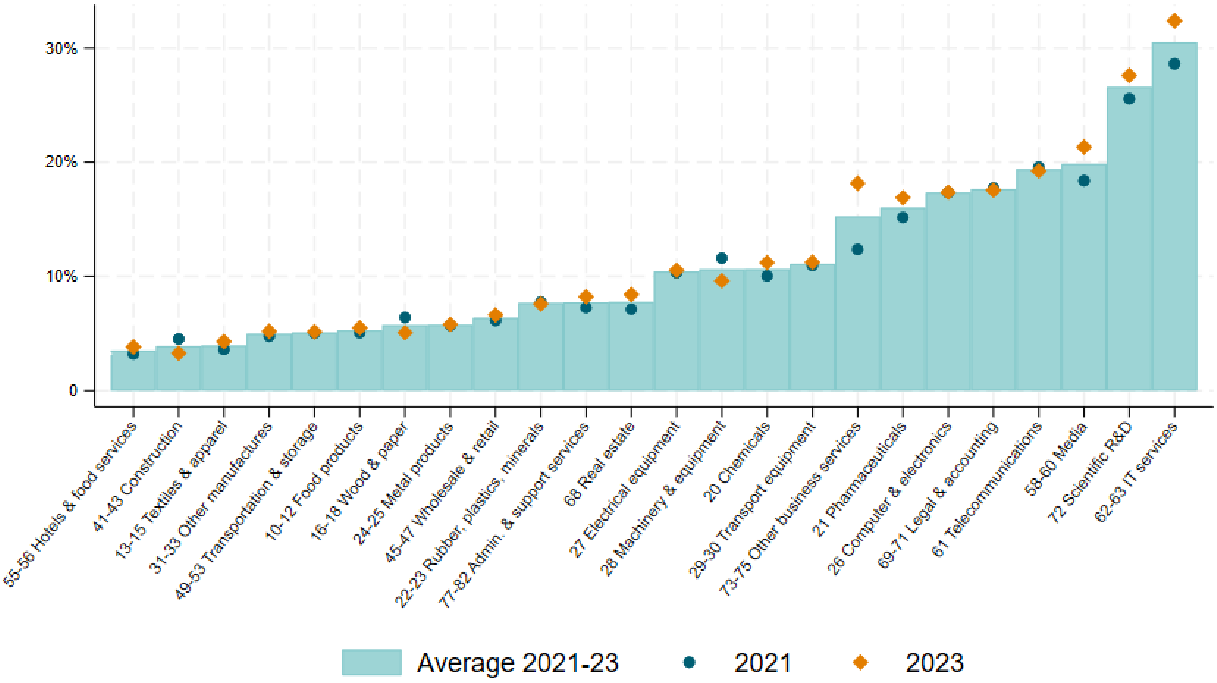
Unlike Figure 2.10, the underlying data for Figure A E.1 in Annex E also encompass non-EU OECD Member countries and hence include information on the ‘Finance & insurance’ (K) sector [see also Figure 3.12 in OECD (2024[43])] and is used to impute the missing information for that sector in the taxonomy and summary indicator presented in the next section.

Some sectors in manufacturing with relatively higher shares of firms using AI are also evident in Figure 2.10, positioning them towards the top half of the sectoral distribution of the AI use indicator. They are manufacturing of ‘Computer & electronics’ and ‘Pharmaceuticals’, where the former also exhibits relatively high AI intensity in the other three indicators considered so far, whereas manufacturing of ‘Pharmaceuticals’ is only highly AI intensive in terms of human capital.

When comparing the shares of AI firms over time in Figure 2.11., AI intensity in many cases remained rather constant or increased from 2021 to 2023. The increases are more evident among sectors that already exhibited higher shares of AI use in 2021, such as ‘Media’ or ‘IT services’, but also in manufacturing of ‘Pharmaceuticals’ as well as ’Chemicals’.

Figure 2.11. Share of firms using AI by sector, 2021 versus 2023

Number of firms using AI technologies as a share of all firms with at least 10 employees, EU27, Norway and Türkiye



Note: This Figure shows the percentage of firms using AI in a given sector and year, which is the number of firms with at least 10 employees using any AI technology relative to all firms of the same size in a specific sector. Only sectors for which the AI intensity taxonomy is constructed are displayed, i.e., manufacturing (excluding coke & petroleum), construction and business services. Sector-specific values are weighted averages over EU27 (Austria, Belgium, Bulgaria, Croatia, Cyprus, Czechia, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Poland, Portugal, Romania, Slovak Republic, Slovenia, Spain, and Sweden), Norway and Türkiye, where the aggregate of EU27 was given a weight of 27/29, and Norway and Türkiye 1/29 each.
Source: Eurostat, https://doi.org/10.2908/ISOC_EB_AIN2

The most substantial increase is observed in ‘Other business services’. These trends are also confirmed in Figure A E.1 in Annex E), showing largest increases from 2020 to 2023 in ‘Information and Communication’ (J) and ‘Professional, scientific & technical activities’ (M) across all OECD Member countries. Although ‘Manufacturing’ (C) exhibits only a modest uptake, this is unsurprising considering its diverse range of production beyond ‘Pharmaceuticals’ and ’Chemicals’. Relevantly, some sectors, such as the manufacturing of ‘Wood & paper’, ‘Coke & petroleum’ as well as ‘Machinery & equipment’ appear to display a decline over time in their share of AI use (see also Figure A E.2 in Annex E).[26] These may be possibly due to specificities in the underlying data confirming the relevance to focus on averages over time while going as granular as possible.

Concerning cross-country variation in sectoral shares of AI use, unreported correlations of the quartile rankings of sectors for each country, for which information is available for all sectors, suggests that the rankings of the AI use indicator reflect, at least to a good extent, sectoral rather than country-specific characteristics.[27]

With the increasing use of large language models (LLMs), which are at the basis of now widely known AI systems, such as ChatGPT©, it appears relevant to expand the scope and explore – to the extent possible – also the use of generative AI technologies. However, while generative AI is evolving at a fast pace and the demand for its application is likely to grow, recent data on the extent to which it is used by firms, including granular sectoral information, are not yet fully available for many countries. One notable exception is Canada, which collected, as part of the 2024 Canadian Survey on Business Conditions, information on firms' use of generative AI. The results mirror the patterns observed for the AI use indicator and show the highest uptake of generative AI in the first quarter of 2024 in 'Information & cultural industries', followed by 'Professional, scientific & technical services' and 'Finance & insurance'. More details are provided in Box 2.4.

Box 2.4. Use of generative AI in Canada

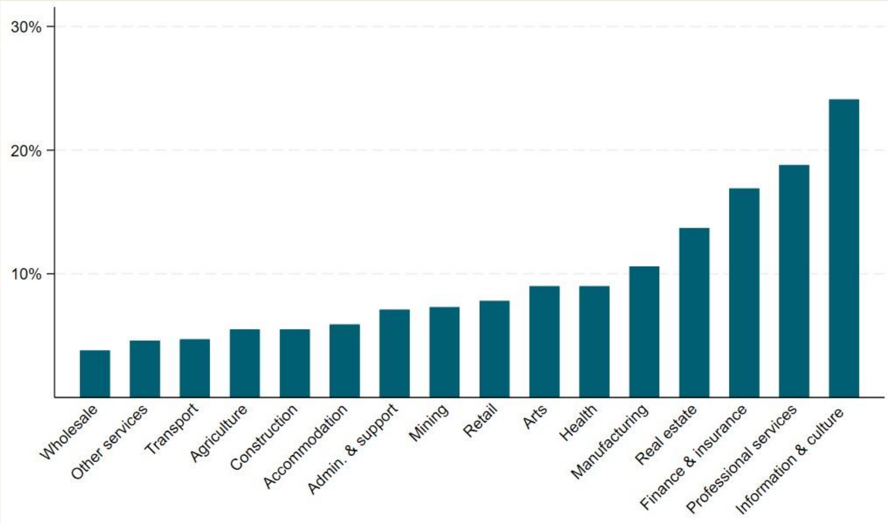
The 2024 Canadian Survey on Business Conditions collected information on business expectations and business conditions in Canada with a component in the first quarter of the questionnaire asking about the business's or organisation's use of generative AI. Generative AI was explicitly defined as follows: "*Generative AI takes data from data scientists and learns to identify patterns based on what it has learned. Examples might include tools like ChatGPT or Dall-E, or other tools that create new or original content, chat responses, designs, or synthetic data.*"

As shown in Figure 2.12, the largest share of firms saying that they are already using generative AI is in business services. In fact, the highest uptake of generative AI is observed in 'Information & cultural industries' with almost one in four firms using it already. This is followed by 'Professional, scientific & technical services' and 'Finance & insurance' with 19% and 17%, respectively, and mirrors the patterns observed for the AI use indicator.

All firms that have already adopted such technology, or have plans to do so, mention accelerating the development of creative content as the main reason to do so. However, motivations vary across sectors: while this appears to be by far the most important driver for firms in 'Professional, scientific and technical services', accelerating the development of creative content and increasing automation in tasks (without reducing employment) are equally important for firms in 'Information & cultural industries'. In 'Finance & insurance', most firms are motivated by increasing task automation and improving client or customer experience. Conversely, in 'Manufacturing' data-driven decision making is most frequently mentioned. The highest shares of firms not having considered to use AI, constituting around 85% in each sector, are found in 'Agriculture, forestry, fishing and hunting', 'Construction' and 'Transportation and warehousing'.

Figure 2.12. Share of firms using generative AI, Canada, Q1 2024

Number of firms already using generative AI as a share of all firms, by NAICS industry



Note: This figure shows the percentage of firms already using generative AI in a given NAICS sector in Q1 2024, which is the number of firms using generative AI relative to all firms in a specific sector.
Source: Statistics Canada, Table 33-10-0784-01 Business's use of Generative AI, Q1 2024, https://doi.org/10.25318/3310078401-eng; and Table 33-10-0785-01 Value or potential value created by Generative AI, Q1 of 2024, https://doi.org/10.25318/3310078501-eng

3 A proposed sectoral taxonomy

This section provides a summary of all indicators related to the different facets of AI -- AI human capital, AI innovation, barrier-adjusted AI exposure, and AI use -- and their intensity at the sectoral level. In particular, Figure 3.1 shows whether each sector has relatively low or high AI intensity based on its ranking distribution for each of the four AI dimensions, described previously. Sectors in the lowest level of AI intensity (bottom quartile) are presented in pale blue, with colour intensity increasing as AI intensity rises, culminating in navy for the most AI intensive sectors (top quartile). Sectors with medium AI intensity (2nd and 3rd quartiles) are shaded in intermediate colours.[28]

While focusing on quartiles provides a relevant summary of the overall patterns observed for the individual indicators, it is only one way to summarise the information. Relevantly, it can nuance the heterogeneity among sectoral AI intensity levels. For instance, in the AI innovation indicator are significant gaps between the first two ('Media' and 'IT services') and the remaining sectors ('Telecommunications', 'Computer & electronics', 'Transportation & storage' and 'Legal & accounting') in the top quartile. Additionally, different quartiles may include sectors with similar levels of AI intensity that are at the extremes of their distributions. This is observed, for example, for the patenting activity in 'Administrative & support services' and the manufacturing of 'Pharmaceuticals', where the share of AI-related patents are 2.2% and 1.1% but they are placed in the second and bottom quartile, respectively. With these caveats in mind, similarities and differences across indicators still emerge.

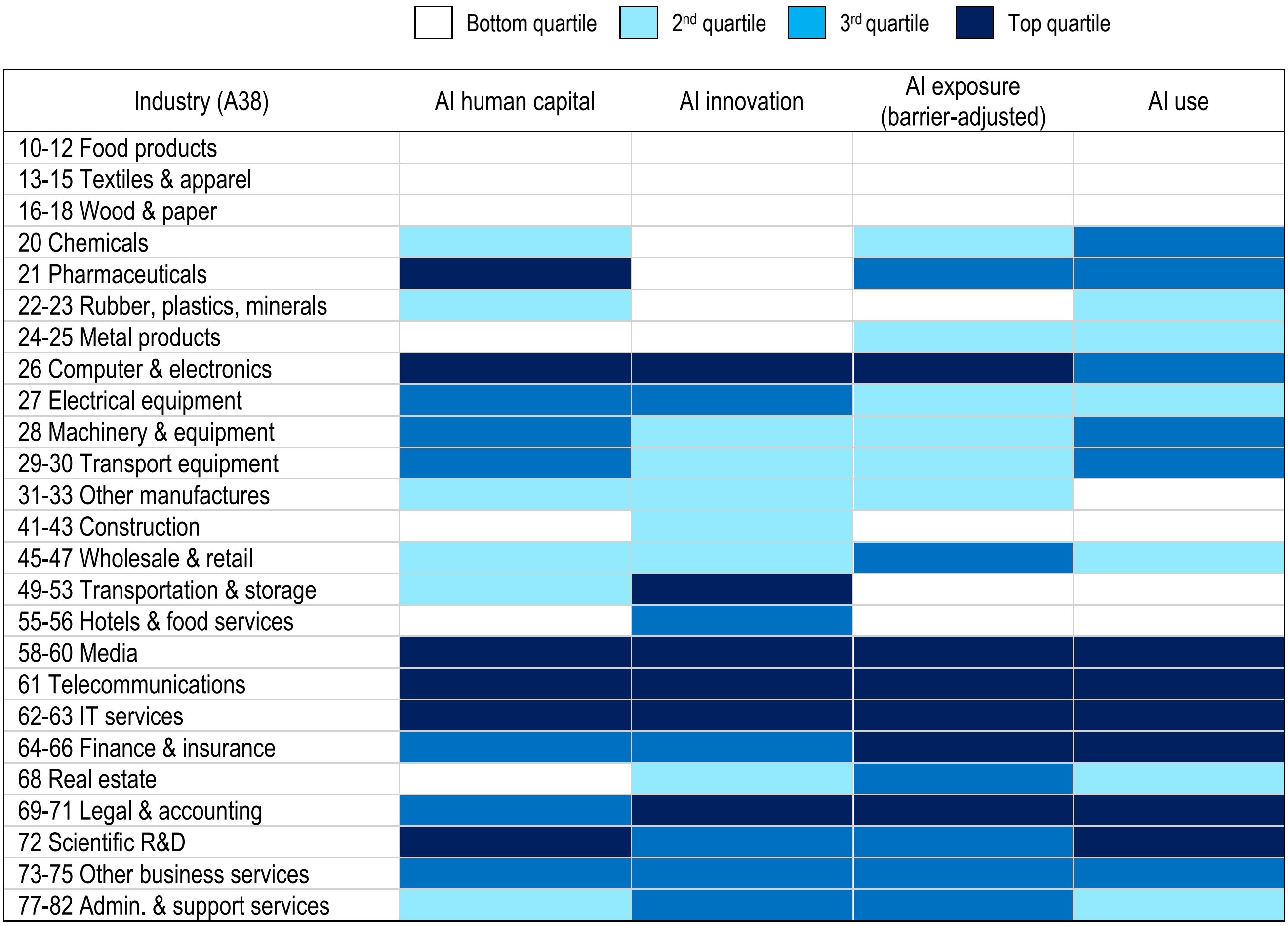
Business services tend to exhibit higher levels of AI intensity in all facets considered. Specifically, 'Media', 'Telecommunications' and 'IT services' rank consistently at the top across all four indicators. Their manufacturing counterpart, i.e., manufacturing of 'Computer & electronics' also displays high levels of AI intensity in all indicators except AI use, where it ranks at the very top of the second quartile. This is followed by 'Finance & insurance', 'Legal & accounting' and 'Scientific R&D', which are in the top quartile of AI intensity in at least two out of the four dimensions.[29] The fact that these sectors rank highly is probably not surprising considering their innovative capacity, reliance on AI use for compliance monitoring and risk assessment, substantial use of data in their business operations, and highly-skilled labour demand.

Conversely, with the exception of 'Computer & electronics' and 'Pharmaceuticals', most manufacturing sectors fall into lower quartiles of the ranking distribution for most AI facets. This includes sectors such as 'Food products', 'Textiles & apparel', 'Wood & paper', 'Rubber, plastics, minerals' and 'Metal products'. Along with 'Construction', these are the least AI intensive sectors across at least two, if not all, indicators considered. This may be the case because of significant investments in legacy infrastructure and equipment that are not easily compatible with AI technologies, in which case human capital and innovative efforts may not be directed towards them yet.

Sectors that are close to the median (i.e., those in the 2nd and 3rd quartiles for different indicators) tend to be characterised by some heterogeneity across indicators. For instance, the 'Electrical equipment' sector scores on the 3rd quartile of AI human capital and AI innovation and on 2nd quartile of barrier-adjusted AI exposure and AI use; the 'Real estate' sector scores in the 3rd quartile of the barrier-adjusted AI exposure, on the 2nd quartile of AI innovation and AI use, and on the bottom quartile of AI human capital. The observed differences may suggest that these sectors are advancing in the diffusion of AI in each facet differently, supporting the multidimensional approach adopted in this work.

Figure 3.1. Sectoral taxonomy of AI intensity, by indicator

Distribution of AI intensity across industries considered, by indicator



Note: The Figure reports the AI intensity of each sector according to each AI indicator (AI human capital, AI innovation, barrier-adjusted AI exposure, and AI use). All underlying indicators are expressed as sectoral intensities, where sectoral values represent averages across countries and years. The colour of the cells in the table corresponds to the quartile of the sectoral distribution to which the sector belongs. Sectors considered are manufacturing (excluding coke & petroleum), construction and business services.
Source: authors' elaboration based on Lightcast, STI Micro-data Lab: Intellectual Property Database, BTOS data, Felten, Raj, and Seamans (2021[4]), Eurostat and OECD Digital Economy Outlook 2024 (OECD, 2024[43]).

Beyond comparing given sectors across indicators, the proposed taxonomy may also enable broader comparisons among the identified facets of AI intensity. While these do not aim to highlight any causal mechanisms and are only based on comparisons of relative positions of sectors rather than actual scores, they may be helpful in uncovering additional policy-relevant insights. In this sense, relating sectoral rankings for the AI human capital or AI innovation indicators to those in the AI use indicator may provide some insights about the potential and the patterns of AI diffusion. For instance, the 'Pharmaceutical' sector scores at the top of the human capital facet but is positioned on the 3rd quartile for AI use, possibly suggesting a potential of AI to further permeate this sector as the demand for specialised human capital translates into relevant AI applications. Relatedly, 'Chemicals' and 'Pharmaceuticals' score at the bottom of the AI innovation indicator but in the 3rd quartile of the AI use one, possibly suggesting that AI use in these sectors may rely more on existing AI applications rather than on pushing the development of novel (and patentable) AI based solutions.

Furthermore, examining sectoral rankings of barrier-adjusted AI exposure alongside actual AI use could also offer insights about the pace of AI diffusion, as it may allow to further relate the potential and actual use of AI based on the relative positions of sectors. For example, the manufacturing of ‘Computer and electronics’ sector ranks highly in barrier-adjusted AI exposure but lower in actual use, possibly suggesting a slower pace of AI diffusion compared to sectors where the relative position in barrier-adjusted exposure is lower than in actual use, such as ‘Chemicals’ or ‘Transport equipment’. These are however sectors that are positioned at the centre of sectoral distributions for both indicators, where sectoral AI intensity levels tend to be more similar.

Towards a possible summary indicator

As previously stated, the information provided by each indicator individually can also be summarised further, in the attempt to provide a simplified operational tool, possibly useful for some follow-up applications. For this purpose, one possible approach is used where each sector is classified as high, medium or low AI intensive in Table 3.1 based on its quartile allocation across the four indicators in Figure 3.1. More specifically, a sector is considered to be highly AI intensive if it ranks in the top quartile for at least two of the four indicators and not in the bottom quartile for the remaining two. Conversely, a sector is deemed low AI intensive if it ranks in the bottom quartile for at least two indicators and not in the top quartile for the remaining two. Sectors falling outside these criteria are classified as medium AI intensive.

The results in Table 3.1 show that overall manufacturing of ‘Computer & electronics’, ‘Media’, ‘Telecommunications’, ‘IT services’, ‘Finance & insurance’, ‘Legal & accounting’ and ‘Scientific R&D’ result being classified as highly AI intensive. Low AI intensive sectors are mainly manufacturing industries, namely ‘Food products’, ‘Textiles & apparel’, ‘Wood & paper’, ‘Rubber, plastics, minerals’ and ‘Metal products’ and ‘Construction’ as well as ‘Hotels & food services’. The other sectors fall in the medium category and include the remaining manufacturing sectors as well as ‘Wholesale & retail’, ‘Transportation & storage’, ‘Real estate’, ‘Other business services’ and ‘Administrative & support services’.

However, this summary indicator compares rather than quantifies AI intensity of sectors and by construction masks some of the heterogeneity observed and discussed in the previous sections, where sectors may score highly in one particular facet but not in others. For instance, ‘Transportation & storage’ is in the top quartile in terms of AI patenting activity but in the bottom quartile in terms of barrier-adjusted AI exposure and AI use, therefore classifying it overall as medium AI intensive. As a result, users may want to carefully consider the purpose of their analysis and consider employing the indicators individually rather than relying on the summary classification.

Table 3.1. Sectoral summary indicator of AI intensity

Summary of AI intensity based on all indicators considered, by sector

Sectoral denomination	Industry (A38)	AI intensity - summary indicator
Food products	10-12	*Low*
Textiles & apparel	13-15	*Low*
Wood & paper	16-18	*Low*
Chemicals	20	Medium
Pharmaceuticals	21	Medium
Rubber, plastics, minerals	22-23	*Low*
Metal products	24-25	*Low*
Computer & electronics	26	**High**
Electrical equipment	27	Medium
Machinery & equipment	28	Medium
Transport equipment	29-30	Medium
Other manufactures	31-33	Medium
Construction	41-43	*Low*
Wholesale & retail	45-47	Medium
Transportation & storage	49-53	Medium
Hotels & food services	55-56	*Low*
Media	58-60	**High**
Telecommunications	61	**High**
IT services	62-63	**High**
Finance & insurance	64-66	**High**
Real estate	68	Medium
Legal & accounting	69-71	**High**
Scientific R&D	72	**High**
Other business services	73-75	Medium
Admin. & support services	77-82	Medium

Note: each sector is classified as high (in bold), medium or low (in italic) AI intensive based on their quartile allocations across the four indicators in Figure 3.1, where a sector is high if it ranks in the top quartile for at least two indicators and not in the bottom quartile for the remaining two, and low if it ranks in the bottom quartile for at least two indicators and not in the top quartile for the remaining two. Sectors falling outside these criteria are classified as medium AI intensive. Sectors considered are manufacturing (excluding coke & petroleum), construction and business services.
Source: authors' elaboration based on Lightcast, STI Micro-data Lab: Intellectual Property Database, BTOS data, Felten, Raj, and Seamans (2021[4]), Eurostat and OECD Digital Economy Outlook 2024 (OECD, 2024[43]).

AI versus digital intensity

There are ongoing discussions concerning the extent to which AI is initiating a new technological paradigm, but it appears that AI systems have certain similarities with other ICTs and IT applications. AI systems indeed rely on and involve the use of computers, computing power, data and other intangibles (Filippucci et al., 2024[44]) that have characterised also previous phases of the digital transformation.

Comparing Table 3.1 above with the analysis carried out by Calvino et al. (2018[1]) (where sectors are classified according to their digital intensity) can provide relevant insights in this context. A summary of some analyses that have utilised Calvino et al.'s (2018[1]) digital taxonomy to assess various aspects, such as market power, firm dynamics, or skill endowment in relation to digital intensity is provided in Box 3.1.

Box 3.1. Taxonomy of digital-intensive sectors

The present study builds upon and significantly extends the taxonomy proposed by Calvino et al. (2018[1]) on the digital intensity of sectors. Both papers highlight that the digital transformation is a complex process that cannot be adequately represented by a single metric and its impact differs significantly across sectors. While the current analysis is centred on AI, Calvino et al. (2018[1]) focused on the digital transformation more broadly, classifying the digital intensity of sectors along different dimensions, including technological components, human capital, and online sales. In particular, the following indicators were used, which also fed into an overall indicator summarising the individual facets of the digital transformation:

- tangible and intangible (i.e. software) ICT investments;
- purchases of intermediate ICT goods and services;
- robots stock;
- ICT specialists; and
- turnover from online sales.

The taxonomy has been used by numerous subsequent analyses, with over 350 citations (as tracked by Google Scholar at the time of writing), to assess the role of digitalisation, for instance, for market power, firm dynamics, or skill endowment, and has been featured in or referenced by several publications in high-level academic journals including. Research Policy (e.g. Domini et al. (2021[45])), Small Business Economics (e.g. Añón Higón and Bonvin (2023[46])), Industrial and Corporate Change (e.g. Reljic, Evangelista and Pianta (2021[47])), Economics of Innovation and New Technology (e.g. Schiersch, Bertschek and Niebel, (2024[48])), The World Economy (e.g. Chiappini and Gaglio (2023[49])), and several others. For example, Calvino and Criscuolo (2019[50]) explored the impact of the digital transformation on business dynamics, focusing on firm entry, exit and job reallocation. They found that digital-intensive sectors, especially digital services, tend to be more dynamic with higher rates of firm entry and job reallocation. However, over the past 20 years, this dynamism has declined in digital-intensive sectors, especially after 2001. Bajgar et al. (2019[51]) focused on the manufacturing sector and used Calvino et al.'s (2018[1]) taxonomy to map the impact of digital technologies on productivity growth, business dynamism, industry concentration, mark-ups as well as mergers and acquisition activities. They observed that while the use of digital technologies is significant, it is not very high and less varied in manufacturing compared to services. Calligaris et al. (2018[52]) found that firm mark-ups are higher in digital-intensive sectors, with the gap between digitally intensive and less-digitally-intensive sectors widening significantly between 2001 and 2014. Berlingieri et al. (2020[53]) examined laggard firms, i.e. firms in the bottom 40% of the productivity distribution, and found younger firms in highly digital industries show potential for faster convergence to the productivity frontier. However, this convergence occurs more slowly, indicating barriers to technology and knowledge diffusion. Other OECD work focusing on skill endowment found that workers in digitally intensive sectors possess higher ICT and advanced numeracy skills (Grundke et al., 2018[54]), especially when employed in occupations that involve tasks contributing to the generation of organisational capital (Cammeraat, Samek and Squicciarini, 2021[55]).

These are only some of the analyses that used such taxonomy, which illustrate the potential of the current study to inform future work aimed at better understanding the role of AI for the economy.

Although AI intensive sectors present a degree of overlapping with digital-intensive sectors, relevant differences emerge, especially when focusing on different facets of AI intensity. Indeed, top AI intensive sectors, scoring high in the summary AI intensity indicator reported above, always result as high or medium-high in the digital intensity summary indicator reported by Calvino et al. (2018[1]).[30] Conversely, not all high-digital-intensive sectors score high in the summary indicator of AI intensity. For instance, manufacturing of 'Transport equipment', while having high digital intensity scores, is overall medium in terms of summary AI intensity. Relatedly, while most of the low AI intensive sectors also have low or medium-low digital intensity, this is not the case for the 'Wood & paper' sector, that has low AI intensity but medium-high digital intensity.[31]

Furthermore, 'Real estate' and 'Transportation & storage' are classified as medium AI intensive in the summary indicator despite their low digital intensity. This difference can possibly be attributed to higher rankings in barrier-adjusted AI exposure and AI innovation, respectively, reflecting a forward-looking approach that considers the future or potential application of AI technologies in these sectors. For example, 'Real estate' may leverage AI for property valuation and predictive analytics, while 'Transportation & storage' could benefit from AI-driven logistics optimisation and innovations in robotics, both underscoring the evolving landscape of AI development and adoption and its transformative potential across diverse sectors.

Relevant differences also emerge for top digital and AI intensive sectors when further considering the different facets of the digital transformation and of AI intensity.[32] For instance, 'Scientific R&D' is both highly digital and AI intensive (when focusing on summary indicators) but differs in terms of human capital requirements. While it is among the top sectors demanding human capital with the necessary AI skills to develop and maintain AI technologies, it falls in the third quartile for the shares of ICT specialists. This suggests that the 'Scientific R&D' sector may require at a fast pace specialised expertise in machine learning, data science, and domain-specific knowledge, despite its relatively lower presence of ICT specialists.

Relatedly, manufacturing of 'Transport equipment' scores in the top quartile for indicators such as the share of turnover from online sales, the presence of ICT specialists as well as robot use, making it a sector with high digital intensity according to the global indicator. However, as mentioned above, it scores in the 2nd or 3rd quartiles for the AI intensity indicators. The 'Wood & paper' sector scores in the 3rd quartile for these same digital intensity indicators (turnover, ICT specialists and robot use), suggesting it may somewhat impacted by the digital transformation. Yet, when assessed through the lens of AI intensity, it falls into the bottom quartile across all indicators, indicating that it is not strongly related to the various facets of AI considered.

Overall, the discussion brings out the relevance of the multidimensional approach adopted in this analysis and highlights that the diffusion and potential of AI, and the resulting AI intensity, appear to build – at least to a certain extent – upon previous phases of the digital transformation. Such complementarities between advanced digital technologies – including AI – and other digital applications or digital infrastructure have been also highlighted by previous OECD work by Calvino and Fontanelli (2023[38]) as well as Calvino, Criscuolo and Ughi (2024[56]), that focus on the role of digital capabilities for AI use and on the relevance of pre-existing complementary technologies for digital technology adoption during COVID-19, respectively.

4 Concluding remarks and next steps

This work has outlined a novel taxonomy of AI intensity, aimed at proxying the extent to which different sectors of economic activity search for AI talent, innovate in the field of AI, are exposed to AI technologies accounting for key barriers, and already use such technologies. For each of these dimensions, the paper discusses the rationale for focusing on them, the indicators used to proxy these and their heterogeneity across sectors, as well as across countries and time when data availability allows.

This taxonomy is a first attempt to provide a more comprehensive picture about AI intensity of sectors, taking into account different facets that are relevant for AI diffusion. It builds upon key indicators that have been used to date, combining them in a consistent framework, highlighting relevant differences in the facets captured by each indicator, accounting for measurement challenges and proposing relevant extensions to the existing literature on the topic. One of the relevant extensions involves the adjustment of a recently proposed indicator of exposure to AI (Felten, Raj and Seamans, 2021[4]), taking into account possible barriers constraining actual exposure and therefore limiting the current potential of AI. This results in a novel indicator of barrier-adjusted exposure to AI, whose comparison with the unadjusted measure may also provide relevant insights for policy makers oriented at boosting AI diffusion or interested in understanding some aspects of the cross-sectoral implications of regulation or ethical considerations.

Key findings highlight significant heterogeneity across sectors and indicators. While some sectors, such as ‘IT services’, ‘Media’ or ‘Telecommunications’, score high along all the dimensions considered, others, such as ‘Transportation and storage’ or ‘Pharmaceuticals’, exhibit more considerable heterogeneity. For instance, the ‘Transportation and storage’ sector shows high AI innovation but low AI exposure and use. The ‘Food products’, ‘Textiles & apparel’ and ‘Wood & paper’ sectors in manufacturing, along with the ‘Construction’ sector, are among the least AI intensive sectors for most indicators considered.

A summary AI intensity indicator has also been reported, based on information resulting from the four indicators, in the attempt to provide a simplified operational tool possibly useful for some follow-up applications. Relating this indicator to the global indicator in the OECD taxonomy of digital intensity (Calvino et al., 2018[1]) suggests that, while AI diffusion appears to rely on the digital transformation, there are also relevant sectoral differences. For instance, not all high-digital-intensive sectors score high in the summary indicator of AI intensity. Relevant differences also emerge when considering further the different dimensions of the digital transformation and AI intensity. For instance, the ‘Computer & electronics’ manufacturing sector ranks at the top in terms of ICT specialists but falls into the third quartile in terms of AI human capital.

The current analysis provides an operational tool to characterise sectors by their AI intensities, focusing on different facets as measured by the different indicators, and hence can be used to explore their correlation with different economic outcomes. As such, the taxonomy complements other established taxonomies, such as those based on R&D or digital intensities.

Such tool appears particularly relevant from a policy perspective, as it can be applied to a range of policy-relevant analyses at the sectoral or at the microeconomic level, e.g., based on firm-level data that contain sectoral information. For instance, future analysis may relate the AI measures proposed or the taxonomy to relevant indicators of business dynamism or to within-sector trends in productivity (e.g., productivity dispersion), allowing the comparison of sectors that have higher or lower AI intensity based on different indicators.

While comprehensive in several respects, the analysis does not aim at capturing *all* possible facets of AI intensity. Future work could possibly extend the number of indicators underpinning the taxonomy, conditional on data availability. In particular, other dimensions of AI intensity of sectors that may be considered by future work could include other proxies of AI innovation, e.g., through the entry of AI startups, of AI use, e.g., focusing on the use of specific AI technologies such as generative AI, or of AI human capital, e.g. through software and computer training necessary to bridge any potential competence gap for AI adoption.

Furthermore, the taxonomy is based on indicators constructed at a moment where generative AI is significantly emerging, while data availability on its diffusion across sectors are still limited. In this sense, continuing to monitor the most recently available data may provide relevant complements to the current taxonomy.

Endnotes

[1] Some indicators are only available for a subset of countries and the precise country coverage for each indicator is elaborated on in section 2, which discusses each indicator in detail.

[2] Further exploring within-sectoral heterogeneity at a more disaggregated level appears challenging considering data availability and measurement challenges (e.g., small cell sizes).

[3] In addition, firms often operate across multiple sectors, and the underlying data sources used for each indicator may classify firms differently, or the primary industry assigned to a firm may not fully capture the breadth of its activities.

[4] This is the case for the adjustment made to the AI exposure where the information on barriers to adoption, obtained from the Business Trends and Outlook Survey (BTOS) and managed by the U.S. Census Bureau, is only available for Manufacturing (NAICS 31-33) as a whole.

[5] Agriculture, utilities, public services, household production and production by international organisations were excluded.

[6] Recent OECD work on the representativeness of Lightcast data observed that they better reflect vacancy distributions than absolute levels, as vacancy shares across sectors and occupations closely follow those in the official data while levels are substantially lower (Tsvetkova et al., 2024[17]).

[7] Aleekseva et al. (2021[9]) and Acemoglu et al. (2022[20]) identify AI jobs when at least one AI skill is evident in the posting, while Borgonovi et al. (2023[13]) classify AI skills into generic (e.g. machine learning or computer vision) and specific (e.g. gradient boosting or natural language processing) ones and identify AI jobs when the vacancy contains at least two generic or one specific skill.

[8] A complementary data source which could be explored in future work is the 2nd Cycle of OECD's Programme for the International Assessment of Adult Competencies (PIAAC), which may allow accounting for software and computer training necessary to bridge any potential competence gap for AI adoption. Alternatively, information on training could be further leveraged, e.g. RISE Israel analysis using LinkedIn data (2024[58]).

[9] The selection of these countries does not come at the expense of generality as is evident when examining the share of AI-related vacancies across broader sectoral categories in other European non-English-speaking countries, with the top three sectors being professional activities, ICT and manufacturing (Figure 3.4 in Borgonovi et al. (2023[13])).

[10] In contrast, Lightcast's data for European countries rely on the European Skills, Competences, Qualifications and Occupations (ESCO) taxonomy covering a lower number of unique skills.

[11] First, AI vacancy shares are averaged across countries (years). Then sectors are ranked each year (country) and divided into four groups according to their rank distribution. Analysing these groups reveals a correlation exceeding 0.8 across time and countries, except for Singapore, which exhibits a lower correlation (>0.6) with other countries.

[12] Future work may further leverage patent data to assess how AI innovation is related to innovations in other fields.

[13] The list of keywords was developed, and the period analysed occurred before the recent boom in generative AI. Hence, related technologies are not included in the innovation indicator. However, it is likely that generative AI technologies tend to be developed in sectors that are at the top of the ranking presented in Figure 2.4, unlikely affecting the relative position of sectors to a major extent. Furthermore, despite relevant heterogeneity – e.g. between core AI patents and industry-specific applications – AI patents are considered, for simplicity, as a single group. Further analysis exploring the heterogeneity of technologies at the core of AI, based on US patent data, is available in Calvino et al. (2023[25]).

[14] For further discussion on the methodology for IP5 patent families, please see Dernis et al. (2015[57]).

[15] Additional robustness analysis focusing on the United States, a country in which patent applications are predominantly filed at the USPTO where patentability requirements for AI-related innovation may be less restrictive (especially when it comes to software-based applications), provide qualitatively similar rankings with respect to those reported in Figure 2.5. In addition, AI patent shares were ranked by industry within each country and compared to the overall sectoral ranking across all countries, which is heavily influenced by US patenting activities, showing a moderate average correlation of 0.6.

[16] Overall, most of these measures tend to positively correlate, as shown by Pizzinelli et al. (2023[35]).

[17] The updated index only considers language modelling applications. The two indices are highly correlated (correlation coefficient: 0.979) (Felten, Raj and Seamans, 2023[33]).

[18] The authors focus on computer vision and find that at current costs US businesses would choose not to automate most vision tasks that have significant "AI Exposure".

[19] The AIIE is the industry equivalent of the AIOE indicator (see Box 2.2 for further details).

[20] In this sense, the BAIIE indicator has a similar rationale to the C-AIOE index by Pizzinelli et al. (2023[35]).

[21] Notably, the sector having the lowest relative ranking has a value for the BAIIE that is equal to the original AIIE value. The sector that has the highest ranking faces the highest downscaling of the AIIE value. See further information in Annex D.

[22] Notice that A38 sector 'Education' results the highest in terms of exposure when considering the full sectoral distribution (see Figure A D.4 in Annex D).

[23] However, it is not known who within firms complete the surveys, particularly given the wide range of topics covered, including firm financials, workforce management and technology adoption. As usual in surveys, the respondents' knowledge and assessment of AI adoption within their firm may impact directly

the indicator's outcome. Furthermore, the indicator does not allow separating out the extent to which AI systems used are embedded in devices or software. Future analysis exploring the role of different types of AI tools may provide additional insights in this direction.

[24] When examining the use patterns of these technologies individually, the sectoral rankings of AI intensity remain stable, except for robotics, where manufacturing industries (despite their relatively low use) move higher in the ranks. Detailed results for each technology and sector are available upon request.

[25] The survey in 2022 did not contain any question about AI use and focused on the use of industrial and service robotics instead.

[26] A notable decrease over time can also be observed in 'Water, sewerage, waste' and 'Construction'.

[27] AI use shares were averaged across the two available years and sectors were ranked (within each country) and divided into four groups according to their rank distribution. Analysing the subset of these 13 countries highlights a correlation exceeding 0.8.

[28] In line with the rest of the analysis, Figure A A.1 in Annex A displays the taxonomy, focusing on all available sectors and including the AIIE index developed by Felten, Raj, and Seamans (2021[4]). Both the AIIE and the BAIIE account for the potential diffusion of AI, but the latter additionally considers how actual AI use barriers may affect this potential.

[29] Figure A A.1 in Annex A shows that, when focusing on all sectors, also 'Education' scores in the top quartile for two out of three indicators available (the AI use indicator is not available), further highlighting the role of AI in this sector.

[30] These comparisons refer to Table 3 in Calvino et al. (2018[1]) and take into consideration the summary (global) indicator of digital intensity focusing on the period 2013-15.

[31] See also further discussion below, focusing on different facets of AI and digital intensity.

[32] These comparisons refer to Table 3.1 above and Table 1 in Calvino et al. (2018[1]).

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Annex A. Additional tables and figures

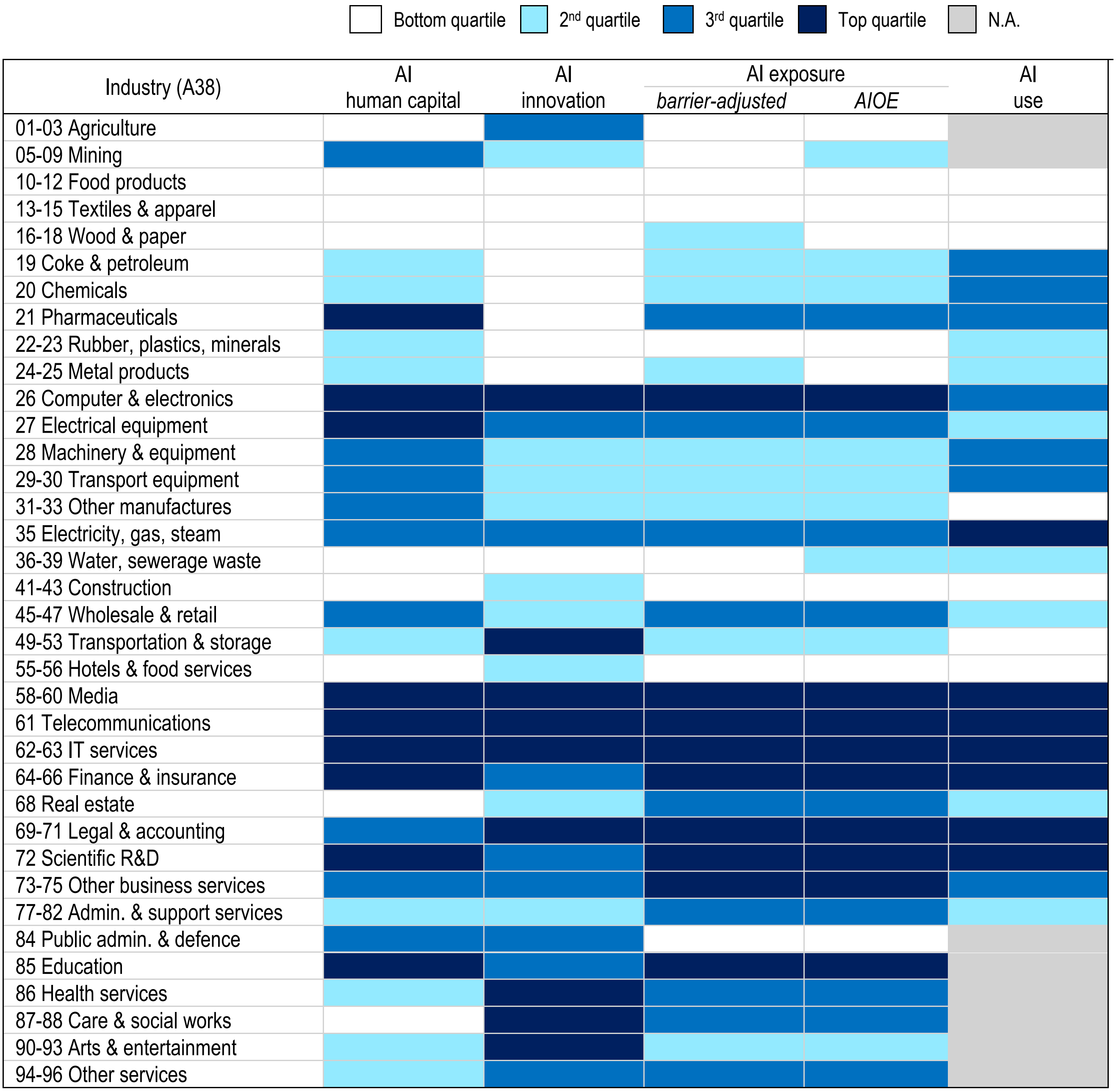
Table A A.1. List of considered sectoral aggregations

Sectoral denomination	Short sector label in Figures	ISIC Rev.4
Food products, beverages and tobacco	Food products	10-12
Textiles, wearing apparel, leather	Textiles & apparel	13-15
Wood and paper products, and printing	Wood & paper	16-18
Chemicals and chemical products	Chemicals	20
Pharmaceutical products	Pharmaceuticals	21
Rubber and plastics products	Rubber, plastics, minerals	22-23
Basic metals and fabricated metal products	Basic metals	24-25
Computer, electronic and optical products	Computers & electronics	26
Electrical equipment	Electrical equipment	27
Machinery and equipment n.e.c.	Machinery	28
Transport equipment	Transport equipment	29-30
Furniture; other manufacturing; repairs of computers	Other manufactures	31-33
Construction	Construction	41-43
Wholesale and retail trade, repair	Wholesale, retail, repairs	45-47
Transportation and storage	Transport services	49-53
Accommodation and food service activities	Hotels & food services	55-56
Publishing, audiovisual and broadcasting	Publishing & broadcasting	58-60
Telecommunications	Telecommunications	61
IT and other information services	IT services	62-63
Finance and insurance	Finance & insurance	64-66
Real estate	Real estate	68
Legal and accounting activities, etc	Law, accountancy & engineering	69-71
Scientific research and development	Scientific R&D	72
Advertising and market research; other business services	Other business services	73-75
Administrative and support service activities	Admin & support services	77-82

Note: Agriculture, forestry, fishing (01-03), Mining and quarrying (05-09), Coke and refined petroleum products (19), Electricity, gas, steam and air cond. (35), Water supply; sewerage, waste management (36-39), Public administration and defence (84), Education (85), Human health activities (86), Residential care and social work activities (87-88), Arts, entertainment and recreation (90-93) and Other service activities (94-96) are excluded.
Source: authors' elaboration.

Figure A A.1. Sectoral taxonomy of AI intensity, by indicator – all available industries

Distribution of AI intensity across industries available, by indicator



Note: The Figure reports the AI intensity of each sector according to each AI indicator (AI human capital, AI innovation, barrier-adjusted AI exposure, and AI use). All underlying indicators are expressed as sectoral intensities, where sectoral values represent averages across countries and years. The colour of the cells in the table corresponds to the quartile of the sectoral distribution to which the sector belongs. This Figure shows all sectors for which data are available.

Source: authors' elaboration based on Lightcast, STI Micro-data Lab: Intellectual Property Database, BTOS data, Felten, Raj, and Seamans (2021[4]), Eurostat and OECD Digital Economy Outlook 2024 (OECD, 2024[43]).

Annex B. AI human capital

Using Lightcast data to identify AI-related vacancies

Lightcast gathers vacancy-related data from over 51 000 online job sites using spider technology, which, on the one hand, continually monitors websites to identify new online vacancies and, on the other hand, extracts job information from a master list of known websites. This two-step process allows Lightcast to collect job postings from various sources, including job boards, government agencies, educational institutions, and numerous employers across different sectors, sizes and locations. The collected data includes company names, occupation, industry information, salary, geographical information, qualifications, and whether the job is full or part-time. Additionally, it provides a list of skills required by employers for each job posting, which encompass over 32 000 unique skills categorised as technical skills, human skills, or certifications.

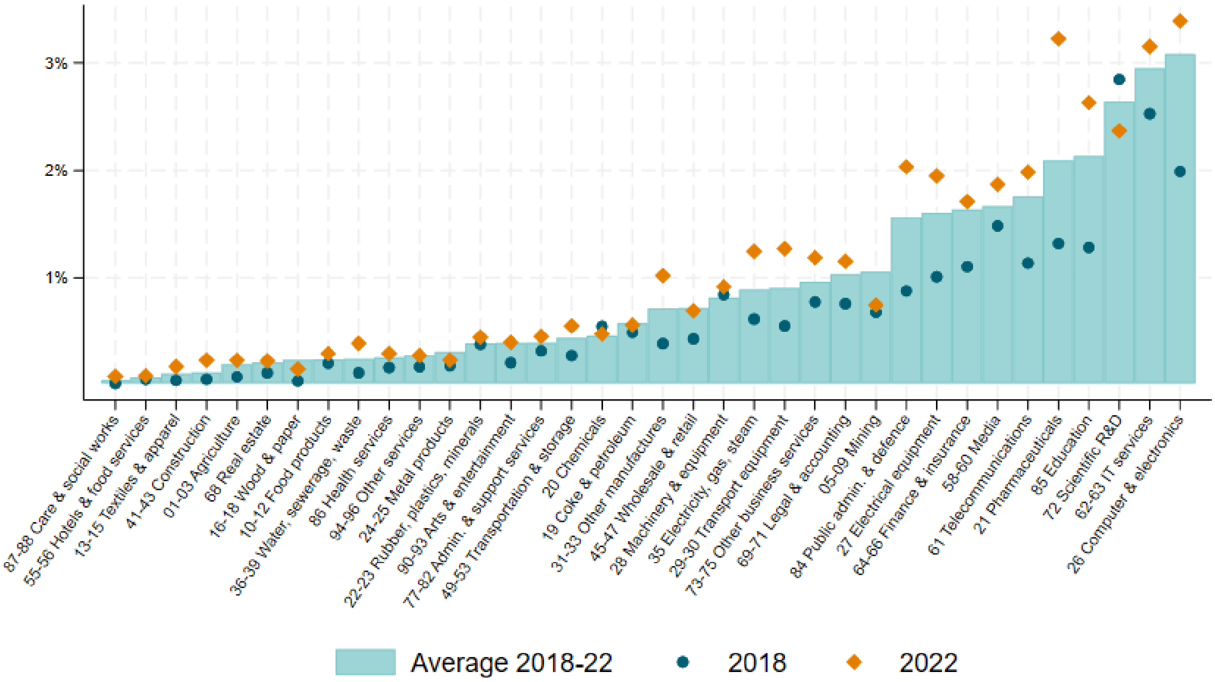
Lightcast also employs a machine learning algorithm to identify duplicates, which may not always be exact copies; the algorithm uses a statistical classifier trained to compare fields such as location, job title, posting text, contact information, and company name, to ensure that each job posting in the dataset is unique.

Online job vacancy data from Lightcast may not fully represent all job vacancies because not all are posted online. However, studies have shown that online vacancies largely mirror trends and industry distributions in official employment data, despite some discrepancies in specific industries and underrepresentation of low-skilled jobs (Cammeraat and Squicciarini, 2021[16]; Vermeulen and Gutierrez Amaros, 2024[18]; Tsvetkova et al., 2024[17]). Therefore, cross-country comparisons should be approached cautiously.

Additional figures

Figure A B.1. Share of AI-related online vacancies by sector, 2018 versus 2022

Number of online vacancies containing AI skills as a share of all vacancies, Canada, Singapore, United Kingdom and United States, all available sectors



Note: This figure shows the percentage of online vacancies containing AI skills in a given sector and year, which is the number of AI-related vacancies relative to all vacancies in a specific sector and year. This Figure shows all sectors for which data are available. Sector-specific values are unweighted averages across countries, namely: Canada, Singapore, United Kingdom and United States. The bar shows the unweighted average across countries and the years 2018-22.

Source: authors' calculations based on Lightcast data.

Annex C. AI innovation

Identifying AI-related patents

The following set of patents are tagged as AI-related and will be used for the aforementioned analysis. This search strategy enables to identify patents that are related to AI, either as they embed core AI techniques (such as specific algorithms or methods) and/or as they relate to technological applications of AI.

- classified in one of the IPC codes: G06N3, G06N5, G06N20, G06F15/18, G06T1/40, G16C20/70, G16B40/20 or G16B40/30
- classified in one of the IPC codes and featuring in their English abstract or claims at least one AI keyword: G01R31/367, G06F17/(20-28, 30), G06F19/24, G06K9/00, G06K9/(46-52, 60-82), G06N7, G06N10, G06N99, G06Q, G06T7/00-20, G10L15, G10L21, G16B40/(00-10), G16H50/20-70, H01M8/04992 or H04N21/466
- classified in one of the CPC codes and featuring in their English abstract or claims at least one AI keyword: A61B5/(7264,7267), B29C2945/76979, B29C66/965, B60G2600/(1876-1879), E21B2041/0028, F02D41/1405, F03D7/046, F05B2270/(707-709), F05D2270/(707-709), F16H2061/(0081-0084), G01N2201/1296, G01N29/4481, G01N33/0034, G01R31/367, G01S7/417, G05B13/(027-029), G05B2219/33002, G05D1/0088, G06F11/(1476,2257,2263), G06F15/18, G06F17/(20-28), G06F19/(34,707), G06F2207/4824, G06K7/1482, G06K9/00, G06K9/(46-52, 60-82), G06N3, G06N5, G06N7, G06N10, G06N20, G06N99, G06Q, G06T2207/(20081,20084), G06T3/4046, G06T7/(00-20), G06T9/002, G08B29/186, G10H2250/(151,311), G10K2210/(3024,3038), G10L15, G10L21, G10L25/30, G11B20/10518, G16B40, G16C20/70, G16H50/(20,70), H01J2237/30427, H01M8/04992, H02P21/0014, H02P23/0018, H03H2017/0208, H03H2222/04, H04L2012/5686, H04L2025/03464, H04L25/(0254,03165), H04L41/16, H04L45/08, H04N21/(4662-4666), H04Q2213/(054,13343,343), H04R25/507, Y10S128/(924-925) or Y10S706
- featuring at least three AI keywords in their English abstract or claims.

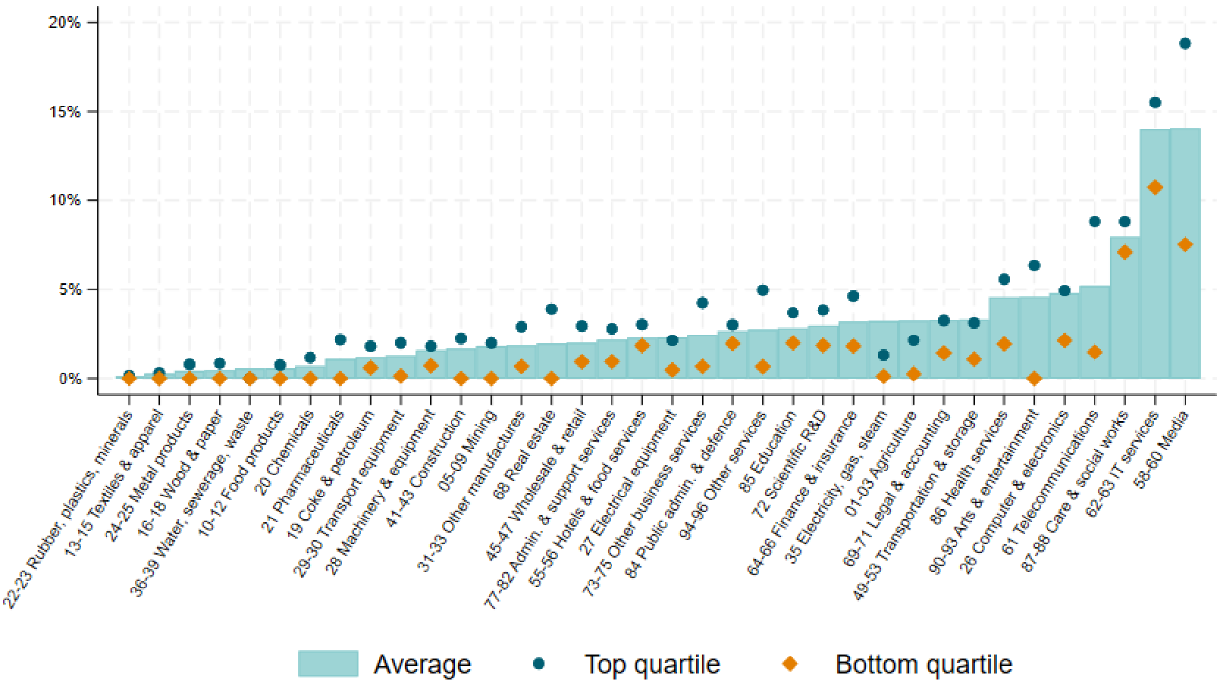
Table A C.1. List of AI keywords and corresponding AI topics

action recognition	genetic algorithm	natural language processing
activity recognition	genetic programming	natural language understanding
adaboost	gesture recognition	nearest neighbour algorithm
adaptive boosting	gradient boosting	neural network
adversarial network	gradient tree boosting	neural turing
ambient intelligence	emotion recognition	neuromorphic computing
ant colony	facial expression recognition	non-negative matrix factorisation
ant colony optimisation	graphical model	object detection
artificial bee colony algorithm	hebbian learning	object recognition
artificial intelligence	hidden Markov model	obstacle avoidance
artificial neural network	hierarchical clustering	particle swarm optimisation
association rule	high-dimensional data	pattern recognition
autoencoder	high-dimensional feature	pedestrian detection
autonomic computing	high-dimensional input	q-learning
autonomous vehicle	high-dimensional space	random field
backpropagation	human activity recognition	random forest
bayesian learning	human action recognition	rankboost
bayesian network	humanoid robot	recommender system
biped robot	human-robot interaction	recurrent neural network
blind signal separation	image classification	regression tree
bootstrap aggregation	image processing	reinforcement learning
brain computer interface	image recognition	relational learning
chatbot	image retrieval	robot
classification tree	image segmentation	rough set
cluster analysis	independent component analysis	rule-based learning
cognitive automation	inductive monitoring	rule learning
cognitive computing	industrial robot	self-organising map
cognitive modelling	instance-based learning	semantic web
collaborative filtering	intelligence augmentation	semi-supervised learning
collision avoidance	intelligent agent	sensor data fusion
community detection	intelligent classifier	sensor fusion
computational intelligence	intelligent infrastructure	sentiment analysis
computer vision	intelligent software agent	service robot
convolutional neural network	kernel learning	similarity learning
cyber physical system	k-means	simultaneous localisation mapping
data mining	latent dirichlet allocation	social robot
decision tree	latent semantic analysis	sparse representation
deep belief network	latent variable	spectral clustering
deep convolutional neural network	learning automata	speech recognition
deep learning	legged robot	speech to text
deep neural network	link prediction	stacked generalisation
dictionary learning	logitboost	statistical relational learning
differential evolution algorithm	long short term memory (LSTM)	stochastic gradient
dimensionality reduction	machine intelligence	supervised learning
dynamic time warping	machine learning	support vector machine
ensemble learning	machine translation	support vector regression
evolutionary algorithm	machine vision	swarm intelligence
evolutionary computation	mapreduce	swarm optimisation
extreme learning machine	markovian	temporal difference learning
face recognition	memetic algorithm	text mining
factorisation machine	meta learning	text to speech
feature engineering	motion planning	topic model
feature extraction	multi-agent system	transfer learning
feature learning	multi-label classification	trajectory planning
feature selection	multi-layer perceptron	trajectory tracking
fuzzy c	multi-objective evolutionary algorithm	unmanned aerial vehicle
fuzzy logic	multi-objective optimisation	unsupervised learning
fuzzy number	multi-sensor fusion	variational inference
fuzzy set	multi-task learning	vector machine
gaussian mixture model	naive bayes classifier	virtual assistant
gaussian process	natural gradient	visual servoing
generative adversarial network	natural language generation	xgboost

Additional figures

Figure A C.1. Country dispersion of AI patents shares by sector, 2017-21

Number of AI patents as a share of total IP5 patents by sector, all available sectors



Note: The bars in the figure shows the percentage of AI patents in a given sector, which is the number of AI patents relative all IP5 patents in a specific sector ("Average"). The dots ("75th percentile") represent the country at the 75th percentile of the distribution in the sector and period considered. The diamonds ("25th percentile") represent the country at the 25th percentile of the distribution in the sector and period considered. This figure shows all sectors for which data are available. Sector-specific values are unweighted averages over the years 2017-21 and countries. Only countries with matching rates to Orbis above 60%, and with more than 15 AI patents in half or more of the relevant sectors in the period considered are included in the figure, namely: Australia, Austria, Belgium, Denmark, Finland, France, Germany, Israel, Italy, Japan, Netherlands, Spain, Sweden, Switzerland, United Kingdom and the United States.
Source: OECD, STI Micro-data Lab: Intellectual Property Database, http://oe.cd/ipstats, March 2024.

Annex D. Barrier-adjusted AI exposure

Data sources

i. AI exposure

The work employs the AI exposure measure at the industry level (AIIE) developed by Felten, Raj, and Seamans (2021[4]) and available online. The data source provides the AIIE measure at the NAICS 4-digit level.

ii. AI barriers

The Business Trends and Outlook Survey (BTOS) – managed by the U.S. Census Bureau – is intended to capture high-frequency changes in economic conditions through a qualitative survey representative of U.S. employer businesses (Bonney et al., 2024[37]). Supplemental content was added to the BTOS from December 2023 to February 2024 to provide detailed information about businesses' use of AI, e.g., in the last six months and six months into the future. Notably, the supplement asks firms that do not expect to use AI in the future why that might be the case, to uncover some of the key impediments to AI adoption or use. The relevant questions used are Question 30 and Question 37.

(Question 30). During the next six months, do you think this business will be using Artificial Intelligence (AI) in producing goods or services? (Examples of AI: machine learning, natural language processing, virtual agents, voice recognition, etc.

1. *Yes*
2. *No*
3. *Do not know*

If 'No' is selected in Question 30, the following question is asked:

(Question 37). Why does this business not plan to use Artificial Intelligence (AI) during the next six months in producing goods or services? Select all that apply.

1. *Too expensive*
2. *AI is not a mature enough technology yet*
3. *Lack of knowledge on the capabilities of AI*
4. *Concerns about privacy/security*
5. *Concerns about bias*
6. *Lack of skilled workforce*
7. *Lack of required data*
8. *Laws and regulations prevent or restrict use of AI*
9. *Previous or current use of AI did not meet expectations*
10. *Other*
11. *AI is not applicable to this business*

The sectoral share of firms reporting a given impediment is obtained by multiplying the sectoral share of firms answering ‘No’ to Question 30 by the share of firms answering the related option in Question 37 (and dividing by 100).

Four barrier categories are then derived: “Applicability”, “Costs and Maturity”, “Operational and Skills”, and “Regulatory and Ethical”. The Applicability barrier includes the option “AI is not applicable to this business”. The Costs and Maturity barrier includes the options “Too expensive” and “AI is not a mature enough technology yet”. The Operational and Skills barrier includes the options “Lack of knowledge on the capabilities of AI”, “Lack of skilled workforce”, “Lack of required data”, and “Previous or current use of AI did not meet expectations”. The Regulatory and Ethical barrier includes the options “Concerns about privacy/security”, “Concerns about bias”, and “Laws and regulations prevent or restrict use of AI”. The “Other” category includes the option “Other”. As firms can select all options that apply, for each barrier category the maximum value across the included options is used, to avoid double counting within barrier categories.

NAICS-2 sectors included in the analysis are Administrative and Support and Waste Management and Remediation Services (NAICS 56), Utilities (NAICS 22), Wholesale Trade (NAICS 42), Agriculture, Forestry, Fishing, and Hunting (NAICS 11), Information (NAICS 51), Finance and Insurance (NAICS 52), Construction (NAICS 23), Retail Trade (NAICS 44-45), Professional, Scientific, and Technical Services (NAICS 54), Manufacturing (NAICS 31-33), Educational Services (NAICS 61), Real Estate and Rental and Leasing (NAICS 53), Transportation and Warehousing (NAICS 48-49), Accommodation and Food Services (NAICS 72), Health Care and Social Assistance (NAICS 62), Arts, Entertainment, and Recreation (NAICS 71), Mining, Quarrying, and Oil and Gas Extraction (NAICS 21), and Other Services (NAICS 81).

Relative barrier indicators

Leveraging the information from the BTOS, a relative sectoral measure of barriers is computed, for each barrier category j (Applicability, Costs and Maturity, Operational and Skills, and Regulatory and Ethical) and sector i.

$\beta_{i,j}$ is defined as the share of firms for category j and sector i reporting barriers (see also above). The relative sectoral measure $\bar{\beta}_{i,j}$ for each sector i within category j is then computed as:

$$\bar{\beta}_{i,j} = \underbrace{\left(\frac{\beta_{i,j} - \beta_{j,MIN}}{\beta_{j,\ MAX}}\right)}_{[0,1)}.$$

$\bar{\beta}_{i,j}$ is the relative position of sector i as compared to other sectors for barrier category j. It is a measure bounded between 0 and 1. The sector i having the lowest relative barrier for category j has a value of $\bar{\beta}_{i,j} = 0$. The sector i having the highest relative barrier for category j has the maximum value of $\bar{\beta}_{i,j}$ across sectors, that is strictly lower than 1.

Figure 2.6 in the main text shows the computation of the relative barrier indicators for “Costs and Maturity”, “Operational and Skills”, and “Regulatory and Ethical”, at the NAICS 2-digit level. Figure A D.1. below shows the computation of the relative barrier indicators for “Applicability”, at the NAICS 2-digit level. The relative indicator for Applicability generally shows less variability across sectors as compared to other barrier indicators.

AIIE adjustment procedure

i. Single barrier-adjusted AI exposure measures ($BAIIE_{i,j}$)

Considering $AIIE_i$, the AI exposure measure by Felten, Raj, and Seamans (2021[4]) for sector i, the (single) barrier-adjusted AI exposure indicator $BAIIE_{i,j}$ for sector i and category j is defined as:

$$BAIIE_{i,j} = AIIE_i \cdot (1 - \bar{\beta}_{i,j}).$$

$BAIIE_{i,j}$ is the (single) barrier-adjusted AI exposure measure for category j, giving the relative position of sector i vis-à-vis other sectors. Adjusted ranking values remain equal to original value $AIIE_i$ if $\bar{\beta}_{i,j}$ is 0, that is if the sector has lowest barrier vis-à-vis other sectors. The adjusted ranking value is downscaled w.r.t. to the original $AIIE_i$ value if $\bar{\beta}_{i,j} > 0$. For the computation at the NAICS 2-digit level, NAICS 4-digit values for $AIIE_i$ are averaged at the corresponding 2-digit level.

The single barrier-adjusted AI exposure measure for "Costs and Maturity", "Operational and Skills", and "Regulatory and Ethical", at the NAICS 2-digit level, is shown in Figure 2.7. Figure A D.2. shows the computation of the single barrier-adjusted AI exposure measure for "Applicability", at the NAICS 2-digit level.

The barrier-adjusted AI exposure measure for "Applicability" closely mimics the unadjusted AI exposure index by Felten, Raj, and Seamans (2021[4]). This is likely because the AI exposure concept is inherently linked to the potential applicability of AI. Conversely, "Costs and Maturity", "Operational and Skills", and "Regulatory and Ethical" barriers may result important mediating factors from potential to actual exposure across businesses and sectors that do not face applicability barriers.[1] In the following and in the main analysis, the focus is thus maintained on "Costs and Maturity", "Operational and Skills", and "Regulatory and Ethical" barriers. Additional analyses are presented below.

ii. Barriers adjusted AI exposure measure ($BAIIE_i$)

The AI exposure adjustment procedure considers the weight $w_{i,j}$ for category j and sector i in terms of total share of firms reporting the barriers as $w_{i,j} = \frac{\beta_{i,j}}{\sum_{k=1}^{3}\beta_{i,k}}$. In particular, $\sum_{k=1}^{3}\beta_{i,k}$ is the sum of the shares for barrier categories Costs and Maturity, Operational and Skills, and Regulatory and Ethical in sector i.[2]

The barrier-adjusted AI exposure indicator $BAIIE_i$, giving the relative position of sector i vis-à-vis other sectors and taking into account the three barrier categories, is defined as:

$$BAIIE_i = AIIE_i \cdot \left[1 - \underbrace{\sum_{j=1}^{3} w_{i,j} \cdot \bar{\beta}_{i,j}}_{\bar{\gamma}_i}\right].$$

Figure A D.3. below reports the $BAIIE_i$ at the NAICS 2-digit level.

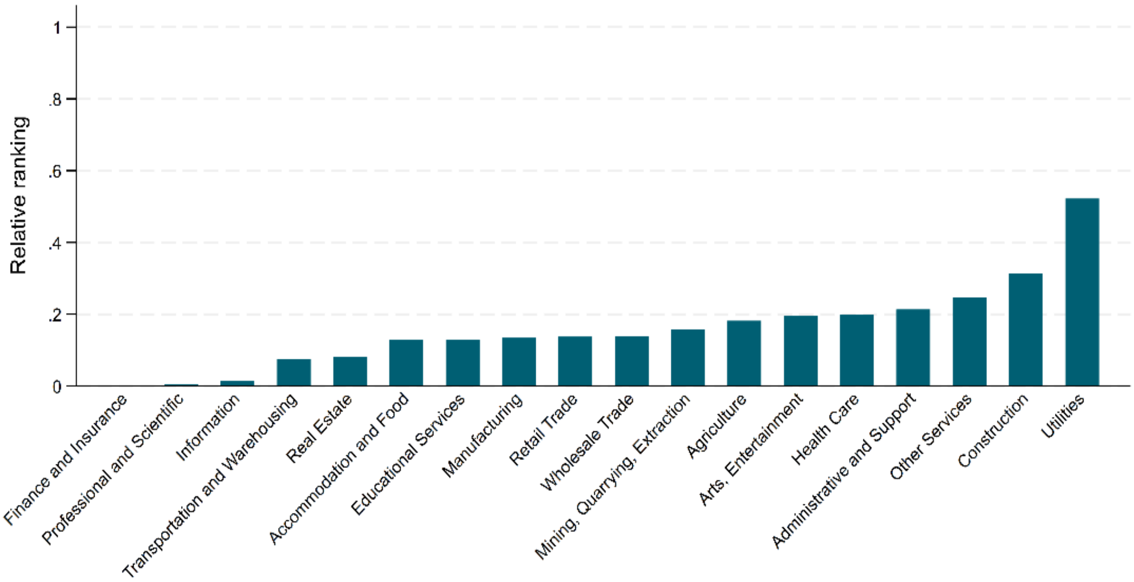
$BAIIE_i$ computation at the STAN A38 level

To compute the $BAIIE_i$ at the STAN A38 level, the following procedure is implemented. The relative weighted barrier indicator $\bar{\gamma}_i$ for NAICS 2-digit sector is imputed at the NAICS 4-digit level. $BAIIE_i$ is then retrieved adjusting the $AIIE_i$ values at the 4-digit level with the imputed $\bar{\gamma}_i$ values, following the equation specified above. Lastly, NAICS 4-digit $BAIIE_i$ sectoral values are aggregated at the STAN A38 level. The most frequent values for STAN A38 are considered for each NAICS 4-digit sector. Values at the A38 level are then retrieved as averages across associated NAICS 4-digit values. The graphs in the main text report the $BAIIE_i$ at the STAN A38 level for the sectors included in the final taxonomy. Figure A D.4. below reports the $BAIIE_i$ at the STAN A38 level for all the available sectors.

Additional results

Figure A D.1. Sectoral Applicability barrier to AI use

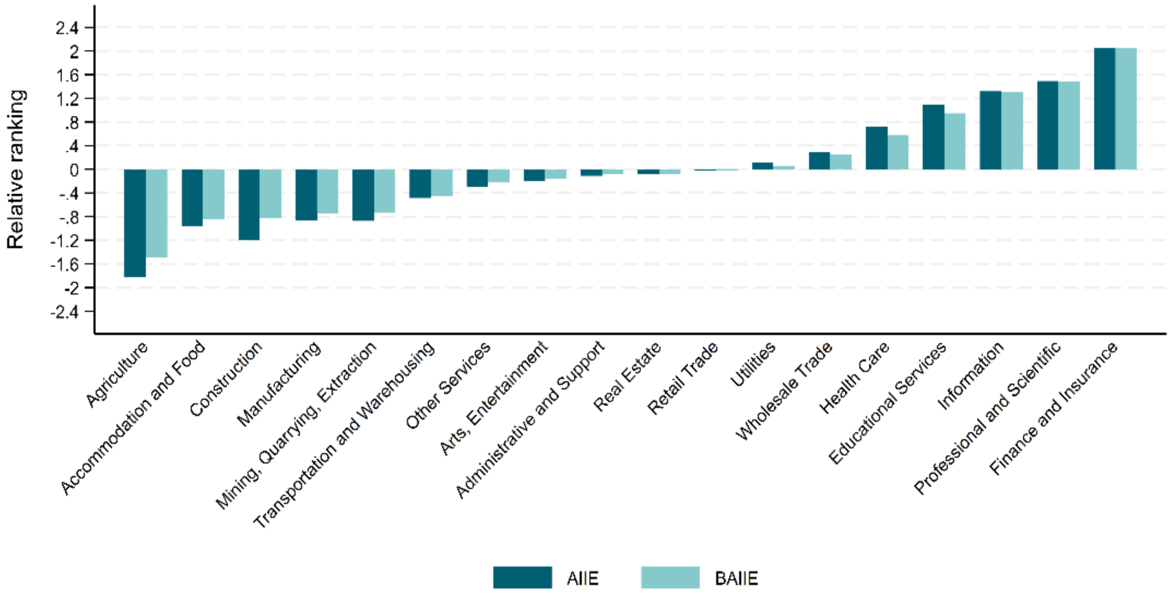
Relative ranking at the sectoral level



Note: The figure shows the relative ranking of NAICS 2-digit sectors according to the “Applicability” barrier. NAICS 2-digit sectors are Agriculture (11, Agriculture, Forestry, Fishing and Hunting); Mining, Quarrying, Extraction (21, Mining, Quarrying, and Oil and Gas Extraction); Utilities (22); Construction (23); Manufacturing (31–33); Wholesale Trade (42); Retail Trade (44–45); Transportation and Warehousing (48–49); Information (51); Finance and Insurance (52); Real Estate (53, Real Estate and Rental and Leasing); Professional and Scientific (54, Professional, Scientific, and Technical Services); Administrative and Support (56, Administrative and Support and Waste Management and Remediation Services); Educational Services (61); Health Care (62, Health Care and Social Assistance); Arts, Entertainment (71, Arts, Entertainment, and Recreation); Accommodation and Food (72, Accommodation and Food Services); Other Services (81, Other Services—except Public Administration).
Source: authors’ calculations based on BTOS data.

Figure A D.2. AI exposure adjusted by Applicability barriers

Relative ranking of sectors according to AIIE and BAIIE measures



Note: The figure reports the AIIE and BAIIE measures at the NAICS 2-digit sectoral level. The BAIIE measure is adjusted according to the "Applicability" barrier. NAICS 2-digit sectors are Agriculture (11, Agriculture, Forestry, Fishing and Hunting); Mining, Quarrying, Extraction (21, Mining, Quarrying, and Oil and Gas Extraction); Utilities (22); Construction (23); Manufacturing (31–33); Wholesale Trade (42); Retail Trade (44–45); Transportation and Warehousing (48–49); Information (51); Finance and Insurance (52); Real Estate (53, Real Estate and Rental and Leasing); Professional and Scientific (54, Professional, Scientific, and Technical Services); Administrative and Support (56, Administrative and Support and Waste Management and Remediation Services); Educational Services (61); Health Care (62, Health Care and Social Assistance); Arts, Entertainment (71, Arts, Entertainment, and Recreation); Accommodation and Food (72, Accommodation and Food Services); Other Services (81, Other Services—except Public Administration). Bars are organised in increasing BAIIE order.
Source: authors' calculations based on BTOS data and Felten, Raj, and Seamans (2021[4]).

Figure A D.3. Barrier-adjusted AI exposure at the NAICS-2 level

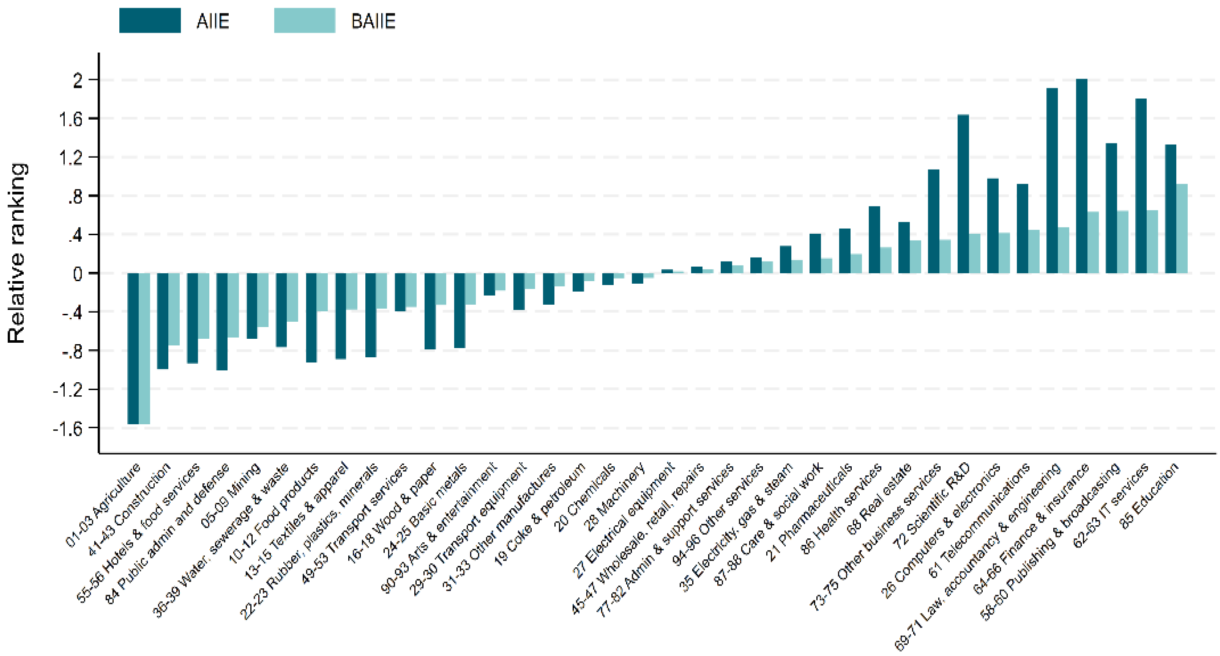
Relative ranking of sectors according to AIIE and BAIIE measures

Note: The figure reports the AIIE and BAIIE measures at the sectoral level. The BAIIE measure adjusts the AIIE index taking into account the role of barriers "Costs and Maturity," "Operational and Skills," and "Regulatory and Ethical". NAICS 2-digit sectors are Agriculture (11, Agriculture, Forestry, Fishing and Hunting); Mining, Quarrying, Extraction (21, Mining, Quarrying, and Oil and Gas Extraction); Utilities (22); Construction (23); Manufacturing (31–33); Wholesale Trade (42); Retail Trade (44–45); Transportation and Warehousing (48–49); Information (51); Finance and Insurance (52); Real Estate (53, Real Estate and Rental and Leasing); Professional and Scientific (54, Professional, Scientific, and Technical Services); Administrative and Support (56, Administrative and Support and Waste Management and Remediation Services); Educational Services (61); Health Care (62, Health Care and Social Assistance); Arts, Entertainment (71, Arts, Entertainment, and Recreation); Accommodation and Food (72, Accommodation and Food Services); Other Services (81, Other Services—except Public Administration). Bars are organised in increasing BAIIE order.

Source: authors' calculations based on BTOS data and Felten, Raj, and Seamans (2021[4]).

Figure A D.4. Barrier-adjusted AI exposure at the A38 level, all sectors

Relative ranking of STAN A38 sectors according to AIIE and BAIIE measures



Note: The figure reports the AIIE and BAIIE measures at the STAN A38 sectoral level. This Figure shows all sectors for which data are available. BAIIE measure adjusts the AIIE index taking into account the role of barriers “Costs and Maturity,” “Operational and Skills,” and “Regulatory and Ethical”. Bars are organised in increasing BAIIE order.
Source: authors’ calculations based on BTOS data and Felten, Raj, and Seamans (2021[4]).

Annex Endnotes

[1] The BTOS publicly available data does not include information on the share of firms selecting multiple options to Question 37. This implies that some firms may select ‘AI is not applicable to this business’ along with other options. However, it appears reasonable to assume that for firms reporting applicability barriers, other forms of barriers are less of a concern and thus not selected.

[2] While the proposed approach is able to adjust the AIIE indicator accounting for the potential role of barriers to AI use, it nonetheless presents some limitations. First, data collected from the BTOS supplement spans a specific period (December 2023 to February 2024) and asks about 6 months in the future. This may not fully capture longer-term trends or shifts in AI use and barriers used to adjust the AIIE measures. Relatedly, the fixed weighting used to aggregate the impacts of different barriers might not accurately reflect the likely dynamic nature of how these barriers influence AI use. In this sense, the analysis presents a static perspective of how AI use barriers may affect actual exposure to AI.

Annex E. AI use

Data source

The AI use indicator leverages the latest publicly available data from the annual Eurostat survey on usage of ICT and e-commerce in enterprises (https://doi.org/10.2908/ISOC_EB_AIN2), consisting of enterprises with 10 or more persons engaged (employees and self-employed persons). In the survey, AI is defined as follows:

Artificial intelligence refers to systems that use technologies such as: text mining, computer vision, speech recognition, natural language generation, machine learning, deep learning to gather and/or use data to predict, recommend or decide, with varying levels of autonomy, the best action to achieve specific goals.

Artificial intelligence systems can be purely software based, e.g.:

- *chatbots and business virtual assistants based on natural language processing;*
- *face recognition systems based on computer vision or speech recognition systems;*
- *machine translation software;*
- *data analysis based on machine learning, etc.*

or embedded in devices, e.g.:

- *autonomous robots for warehouse automation or production assembly works;*
- *autonomous drones for production surveillance or parcel handling, etc.*

Notably, the module (F in 2021 and E in 2023) on AI is only asked to enterprises with access to the internet (question 1 in module A in 2021 and 2023) and is as follows:

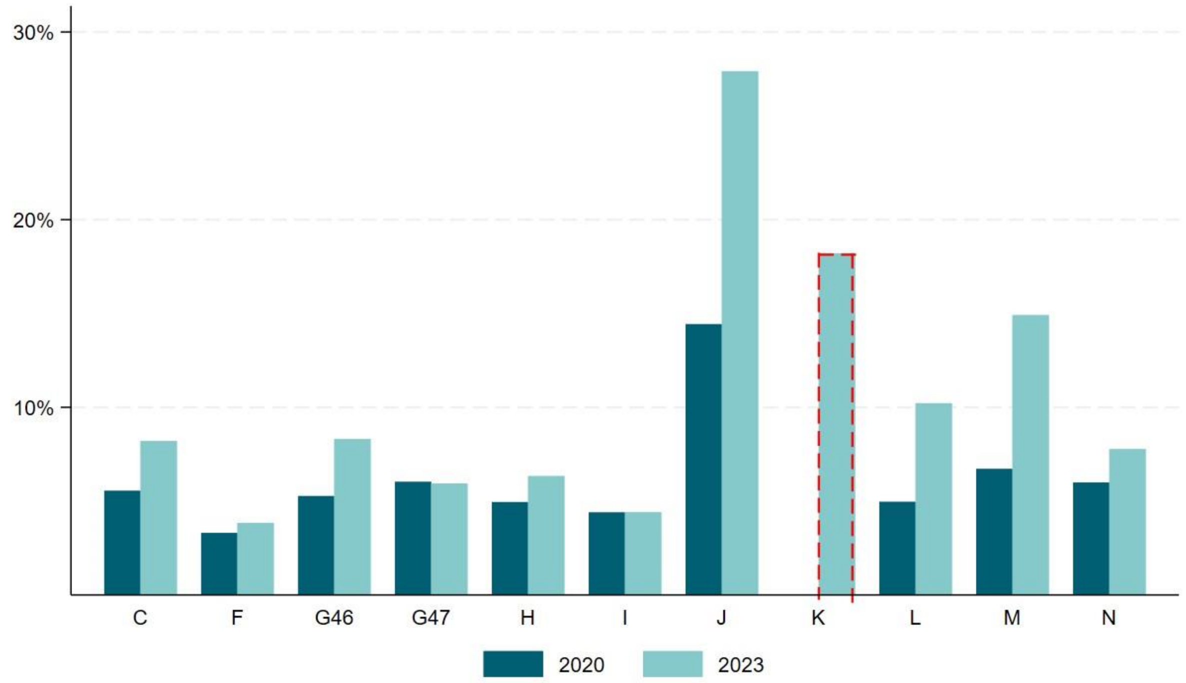
Does your enterprise use any of the following Artificial Intelligence (AI) technologies?

1. *AI technologies performing analysis of written language (e.g. text mining)*
2. *AI Technologies converting spoken language into machine-readable format (speech recognition)*
3. *AI Technologies generating written or spoken language (natural language generation, speech synthesis)*
4. *AI Technologies identifying objects or persons based on images or videos (image recognition, image processing)*
5. *Machine learning (e.g. deep learning) for data analysis*
6. *AI Technologies automating different workflows or assisting in decision making (e.g. AI based software robotic process automation)*
7. *AI Technologies enabling physical movement of machines via autonomous decisions based on observation of surrounding*

Additional figures

Figure A E.1. Share of firms using AI by sector, 2020 versus 2023

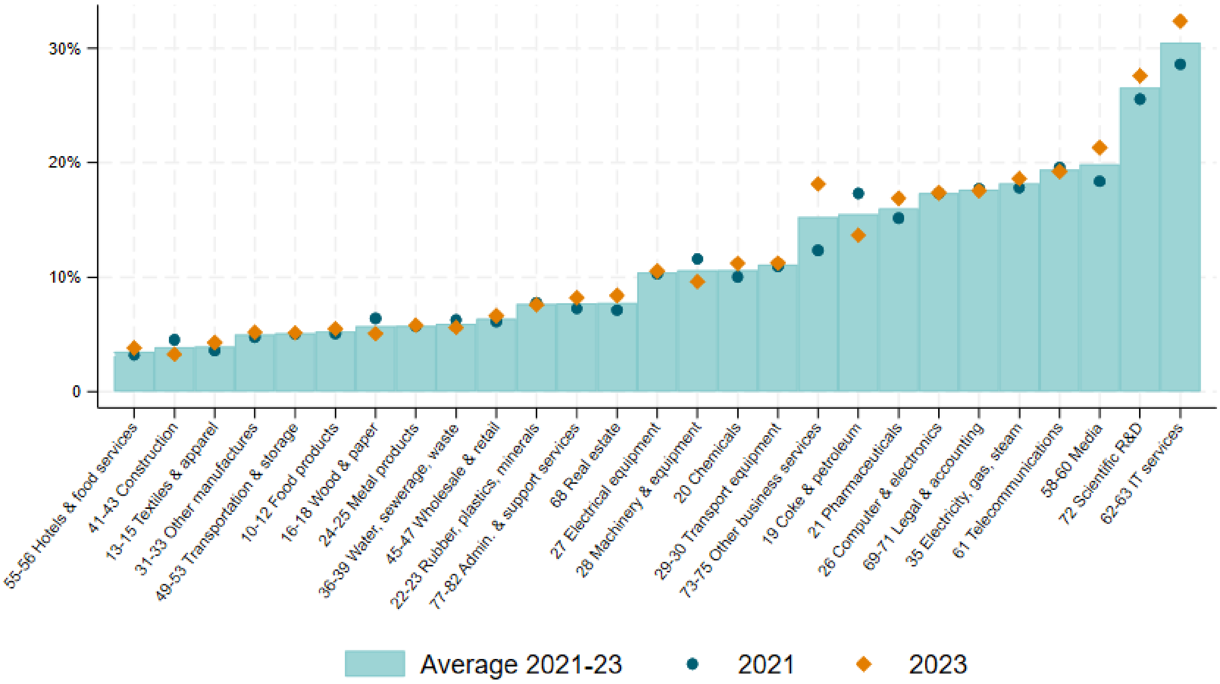
Number of firms using AI technologies as a share of all firms with at least 10 employees, OECD countries



Note: This figure shows the percentage of firms using AI in a given sector and year, which is the number of firms with at least 10 employees using any AI technology relative to all firms of the same size in a specific sector. This Figure shows all sectors for which data is available. Data for OECD countries that are part of the European Statistical system are obtained from the original indicator as published by Eurostat (see Figure 2.10). Data for Australia, Brazil, Canada, Colombia, Japan, Korea, Mexico, New Zealand, Switzerland and the United States were collected by the OECD. The bar for K, highlighted in red, represents the share of AI using firms for only the latter countries and was obtained from Figure 3.12 of the OECD Digital Economy Outlook 2024 (OECD, 2024[43]).
Source: OECD's ICT Access and Usage by Businesses database.

Figure A E.2. Share of firms using AI by sector, 2021 versus 2023

Number of firms using AI technologies as a share of all firms with at least 10 employees, EU27, Norway and Türkiye, all sectors



Note: This figure shows the percentage of firms using AI in a given sector and year, which is the number of firms with at least 10 employees using any AI technology relative to all firms of the same size in a specific sector. This Figure shows all sectors for which data are available. Sector-specific values are weighted averages over EU27 (Austria, Belgium, Bulgaria, Croatia, Cyprus, Czechia, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Poland, Portugal, Romania, Slovak Republic, Slovenia, Spain, and Sweden), Norway and Türkiye, where the aggregate of EU27 was given a weight of 27/29, and Norway and Türkiye 1/29 each.
Source: Eurostat, https://doi.org/10.2908/ISOC_EB_AIN2